1 GENERAL

1.1 SCOPE

The purpose of this document is to cover the design, detail engineering, procurement and supply of material, fabrication, assembly, expediting and inspection, testing, painting, packing, marking, guarantees, performances, documentation, certification and permits of the following API 610 Type medium duty process centrifugal Pumps:

BASE CASE to be included in the base scope of supply:

Equipment TAG	Equipment Description	N. Required	Quality Code	Area Classification
51-P-1003 A/B/C/D	Pumparound Pump	4	2	Zone 2, IIA T3 (HOLD)
51-P-1004 A/S	Passivation Pump	2	2	Zone 2, IIA T2 (HOLD)
51-P-1006 A/B	Reactor Effluent Pump	2	2	Zone 2, IIA T3 (HOLD)
51-P-2002 A/B	Recycle Column Reflux Pump	2	2	Zone 2, IIA T3 (HOLD)
51-P-2004 A/B	Butene-1 Column Reflux Pump	2	2	Zone 2, IIA T3 (HOLD)
51-P-2006 A/B	C5+ Product Drum Pump	2	2	Zone 2, IIA T3 (HOLD)
51-P-2007 A/B/C	Butene-1 Product Pump	3	2	Zone 2, IIA T3 (HOLD)
56-P-7001 A/B	HP Flare Knock-Out Drum Pump	2	2	Zone 2, IIA T3 (HOLD)
NOTE: For all items mentioned above Instruments shall be certified as minimum for Zone 1 IIC T3				

Including all equipment and ancillaries as hereinafter specified, in accordance with the requirements set forth in this Supply Specification and referenced documents.

The above listed equipment will be installed in the “First Integrated Gas Chemical Complex. Second phase (Polyethylene production)” in the Karabatan, Kazakhstan, located 32 km to the north-east of Atyrau city on the territory of the special economic zone (SEZ) as part of the SILLENO Project.

The Equipment part of this Supply Specification shall be designed for a service life of 25 years and a minimum site turnaround interval of 5 years.

1.2 TERMS AND DEFINITIONS

The following terms used in this document have the meaning defined below:

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- MR Material Requisition
- NPS Nominal Pipe Size
- P&ID Piping & Instrumentation Diagram
- PIM Pre-Inspection Meeting
- PO Purchase Order
- SAT Site Acceptance Test
- SIF Safety Instrumented Function
- SIS Safety Instrumented System
- SI International System of Units
- SIL Safety Integrity Level
- SS Supply Specification
- TM Tecnimont Standards
- VSD Variable Speed Drive

1.3 ORDER OF PRECEDENCE

In case of conflict among the requirement of this specification and other documents applicable to this project, **VENDOR** shall consider the following rank of precedence:

- ✓ Licensor standards (where foreseen)
- ✓ Local laws and regulations
- ✓ Agreed Technical Deviations
- ✓ The Material Requisition
- ✓ Project General Specifications
- ✓ Project Standards
- ✓ International Codes and Standard.
- ✓ Other referenced documentation

As a general guideline, the most stringent requirements will govern. However, **VENDOR** shall inform **CONTRACTOR** about the discovered conflict to solve the inconsistency.

Conflict resolution and / or interpretation, based on mentioned order of precedence, shall be obtained from **CONTRACTOR** in writing for **VENDOR** to proceed with relevant due action.

CONTRACTOR will not be held responsible of any consequence out-coming from misinterpretation, **VENDOR**'s standard data proposed by **VENDOR** on revised technical offer or **VENDOR**'s filled in data table (.xls file) tabulation, technical clarification, signed MR documents etc which after issuance of PO found non-compliance or inconsistent within various project specifications attached with MR or non-compliance with above indicated instructions, there including any action should be required on **VENDOR**'s side to fully comply with the contractual obligations without commercial impact on **CONTRACTOR/OWNER.**

1.4 INSTRUCTION TO BIDDERS

The **BIDDER** shall submit a proposal to meet the requirements set forth in this specification including its attachments and in accordance with the applicable local Regulations, Codes and Standards indicated in Project Specifications referenced into this Supply Specification.

During bid review stage, **BIDDER** might be required to attend one or more Technical Clarification

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meeting at **CONTRACTOR** offices to clarify unclear matters related to his proposal and mutually define the scope of supply.

After or during technical clarification phase, **BIDDER** may be requested to issue a revised technical proposal.

Material Requisition, this specification and the mechanical data sheet will be marked up by **CONTRACTOR** according to the latest technical proposal as reviewed and finally agreed between **CONTRACTOR** and **BIDDER**.

BIDDER shall be requested to sign a copy of the above marked documents on each page for acceptance.

The **BIDDER** shall attach to his proposal as minimum the following technical information and deliverables:

- Filled in Mechanical Data Sheet
- Preliminary Layout and static loads
- Preliminary General Arrangement (for item installed on concrete pedestal, a preliminary pedestal dimension and footprint shall be available along with GA drawing)
- Detailed Deviation List
- All documents required with bid shown in Attachment 2 “Vendor Required Documents List (VRDL)” of this supply specification

1.5 REFERENCES LIST

BIDDER shall submit with proposal a reference list demonstrating his experience in manufacturing proposed equipment under similar operating/design conditions.

The **BIDDER** shall have manufactured at least 2 similar equipment running under similar service conditions over a period of at least five (5) years demonstrating good operation and reliability. Prototype design/machines are not acceptable. 2

1.6 ALTERNATIVE DESIGN

Base quotation shall be based on the prescriptions of this Supply Specification. Additionally, **BIDDER** can propose alternative solutions, more reliable from technical perspective and more competitive from the commercial point of view and without any reduction of the supply quality. **BIDDER** shall not take into consideration alternate proposal unless approved by **LICENSOR/CONTRACTOR/OWNER.**

1.7 CLARIFICATIONS AND DEVIATIONS

BIDDER shall submit a base proposal complying strictly with this requisition and all applicable documents.

Any technical deviations from the bid inquiry, purchase order and its attachments, including, but not limited to, the data sheets and narrative technical specifications shall be sought by the **BIDDER** only through Clarifications/Deviations sheet format provided by **CONTRACTOR**.

BIDDER is advised that clarifications, deviations and exceptions considered by **CONTRACTOR** are only those separately specified in the proposed Deviation/Alternative/Clarification Sheet accompanying **BIDDER** proposal. **CONTRACTOR** has no responsibility to discover any deviations, exceptions or

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alternatives entered in other parts of **BIDDER** proposal than in the aforesaid Deviation/Alternative/Clarification Sheet.

If in the experience/knowledge of **VENDOR** and/or his **SUB-VENDOR's**, any requirements of this specification create, even potentially, unsafe (for personnel and/or plant) and/or less reliable design for the project (during start-up/normal/emergency operations), it is **VENDOR**'s responsibility to bring, in writing, such situations to the attention of the **CONTRACTOR**. If no situations have been identified in writing, it is understood that equipment and services will be provided as per the specification with sole responsibility resting with **VENDOR** for safe and reliable operation of the plant during various operating and environmental conditions throughout the plant life.

Concession requests after PO assignment require **CONTRACTOR**'s and **OWNER**'s review/approval, prior to the proposed technical changes being implemented. Technical changes implemented prior to **CONTRACTOR/OWNER** approval are subject to rejection.

The deviations shall not affect the safety, reliability & performance of the equipment.

If the Bid is fully compliant, **BIDDER** shall submit the Clarifications/Deviations/Exceptions sheet format with marked **"No deviations"** in the Deviation column and sign it.

No deviations shall be allowed from authority requirements.

1.8 COMPLIANCE WITH SPECIFICATIONS

After order award, the **CONTRACTOR** will not allow any price and/or delivery time increase, necessitated from failure to comply with this Supply Specification and relevant Attachments, Annexes and Reference Documents.

1.9 VENDOR STATEMENT

With the signature of the Material Requisition, the present Supply Specification and related Mechanical Data Sheet(s), **VENDOR** accepts that after Purchase Order, no reference shall be made to **VENDOR** proposal (Bid).

The Material Requisition signed by the **VENDOR** shall be the only technical document valid after P.O.

2 REFERENCE DOCUMENTS

VENDOR shall refer, in development of its activities, to all the documents listed in Attachment 1, in any kind of format (paper or electronic format) or status, in full content and purpose for which they were issued.

If a document listed has not been provided, either in request for quotation or at Purchase Order receipt, it is **VENDOR**'s responsibility to request it and ensure that equipment/design complies with it.

3 CODES AND STANDARDS

Applicable Codes, Standards and Practices, are those referenced within this Supply Specification and all other Specification, Mechanical Data Sheet, Technical Specification, Project Standard, Engineering Drawing therein listed. The latest edition at the Purchase Order date of the applicable laws, regulations, codes, standards and practices, shall be adhered to.

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It shall be the **VENDOR**'s and his **SUB-VENDOR'**s responsibility to acquire possession of all required codes and standards referenced.

4 ENVIRONMENTAL DESIGN DATA

4.1 MEASURING SYSTEM

Units to be applied on all Project documents shall be International System of Units (SI) as shown in 4503-YZ-SG-T0000001 (SLN-2725-0000-00-A-0001) "Basic Engineering Design Data"

4.2 LANGUAGE

VENDOR shall use English language for all correspondence and communications and in general for the execution of the Purchase Order.

All drawings and documents submitted by **VENDOR** for **CONTRACTOR'**s review and approval shall be in English Language till the final approval. Once the drawings and documents will be approved as final with no comments, they must be translated also in Russian language.

IOM, TR CU and Technical Passports must be in English, Russian and Kazakh language as required by the local rules.

4.3 SITE CONDITIONS

For site environment conditions **VENDOR** shall refer to document 4503-YZ-SG-T0000001 (SLN-2725-0000-00-A-0001) "Basic Engineering Design Data".

The mechanical equipment and components will be installed outdoor, under shelter and shall be designed for the specified outdoor design conditions.

MDMT = -38°C.

4.4 WINTERIZATION/TROPICALIZATION

Winterization/Tropicalization shall be provided for all fluids that can potentially freeze/overheat during cold/hot ambient temperatures, including utilities.

VENDOR shall consider the following features for winterization/tropicalization:

- equipment suitable for site minimum/maximum design temperature.
- appropriate oil system heat-tracing/cold/hot weather protection, if applicable.
- appropriate shaft seal system heat-tracing/cold/hot weather protection, if applicable.
- heat/cold tracing and insulation to prevent icing/overheat of all draining facilities from equipment handling fluid prone to freeze.
- heat/Cold tracing and insulation of equipment including auxiliary piping on the skid, if applicable.
- instruments suitable for low/high ambient temperature.
- heating Box / sun shelter for the instrument itself and the hook up.
- Heat/Cold tracing and insulation for instrumentation and hook up, low points and dead legs where water and/or liquids prone to freeze can settle.

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4.5 UTILITIES CONDITIONS

For available utilities characteristics at the equipment battery limit refer to document 4503-YZ-SG-T0000002 (SLN-2725-0000-00-A-0002) "Utility and Auxiliary Materials Conditions and Specifications".

Utility conditions indicated in above mentioned document are at plant battery limit. Correct value available at package location must be discussed during technical alignment.

Utilities like Instrument Air, Plant Air and Nitrogen could reach the package inlet with a minimum temperature of minus 38 °C.

Vendor must identify and list the items/parts/skids which do not withstand minus 38 °C due to mechanical reasons. When items are identified, minimum acceptable temperature shall be defined for each of these items/parts/skids.

Vendor must check if a suitable solution (like online electrical heaters to be installed at the inlet of each skid for each utility) can be provided and quote it as option.

5 HEALTH, SAFETY AND ENVIRONMENT (H.S.E.)

Design studies, material supply and construction activities shall comply with all the H.S.E aspects of the project. **VENDOR** shall demonstrate his own H.S.E commitment (H.S.E policy, Design HSE Plan).

VENDOR shall ensure that the design, fabrication, and testing of his equipment does not endanger the health and safety of his own employees, employees of his **SUB-VENDOR**, employees of site erection **CONTRACTOR**, and future operators of the Plant.

VENDOR shall provide necessary recommendations for a safe installation, operation, and maintenance of his equipment. Equipment emissions shall comply with the local environmental regulations.

VENDOR shall be responsible for ensuring that equipment/package meet all applicable regulations on health, safety, and environmental issues.

HAZOP (HAZard and OPerability) study and SIL (Safety Integrity Level) classification reviews will be performed during detailed engineering phase.

Upon **CONTRACTOR**'s or **OWNER'**s decision, **VENDOR** may be required to participate to such sessions, which shall be part of his contractual obligations.

HAZOP & SIL Review participation & consequent implementation of the implications shall be in **VENDOR**'s scope. No cost and time implication shall be entertained by **VENDOR** in relation to HAZOP/SIL result after order placement.

6 SCOPE OF SUPPLY

6.1 SUPPLY INCLUSIONS

VENDOR's scope of supply shall include, but not be limited to, the following:

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Sr. No.		Description	Included in Scope	
			Yes / No / NA	Remarks
		ITEM DETAILS – MEDIUM DUTY CENTRIFUGAL PUMPS		
1.	☒	Medium duty centrifugal Pumps as per API 610 latest edition and other applicable project specifications, along with Drive Motor, associated auxiliaries (Coupling, Seal System, Interconnecting Piping, Electrical & Instrumentation items as necessary).		
2.	☒	Fixed speed electrical motor as necessary depending upon pump type (Sized for end of curve operation when mentioned in datasheet) and motor terminal boxes, complete with motor mounting nuts & bolts for mounting on common baseplate with main pumps.		
3.	☒	Mechanical seals and shaft sealing system (Seal Flushing and Cooling system), in accordance with respective mechanical datasheet indications, with complete auxiliaries including Seal Pot, cooler, Interconnecting Piping, valves, fittings, gaskets, bolting, supports, instruments, and any other accessories required as per API 682 latest edition & project specification and for proper and reliable operation of the equipment.		
4.	☒	Lubrication requirement as mentioned in respective data sheet.		
5.	☒	Constant level oiler on each bearing housing.		
6.	☒	Flexible element disk coupling, spacer type, non-lubricated couplings shall be furnished for all pumps. API 671 coupling to be provided when mentioned in respective pump datasheets / applicable specification.		
7.	☒	Non sparking coupling guard totally enclosing coupling and shaft end.		
8.	☒	Torsional study including report, if applicable.		
9.	☒	Lateral study including report, if applicable.		
10.	☒	Post Weld Heat Treatment (PWHT) for all fabricated components.		

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Sr. No.		Description	Included in Scope	
			Yes / No / NA	Remarks
11.	☒	Common baseplate/mounting plate for pump, driver and seal system. Complete single skid unit for mounting of all the items and components included in **VENDOR's** scope of supply. In case all the items and components cannot be mounted on a single skid then **VENDOR** shall advice the no. of multiple skids with dimensions.		
12.	☒	Any other equipment required as per **VENDOR** recommendation for smooth & satisfactory operation of API 610 Medium duty Centrifugal Pumps.		
13.	☒	**Piping (Within Skid/ Interconnecting between skids):** a. Interconnecting piping within the skids with fittings, valves (isolation valves, non-return valves etc.), safety / relief valves, control valves, gaskets and bolting. Prefabricated interconnecting piping between skids shall also be supplied by the VENDOR. All loose supplied piping will be prefabricated, pickled, covered with primer, painted and tested up to final coat. b. All interconnecting piping shall be supplied according to Project Pipe class referenced in **Attachment 1**. Any missing pipe class shall be requested to CONTRACTOR by VENDOR. c. Design of supports for interconnecting piping and supply of primary support; secondary support by CONTRACTOR but Vendor shall provide loads to allow CONTRACTOR designing the support. d. Drains and vents shall be Valved / blinded, extended up to edge of base plate / package battery limit (common header with single tie-in point) e. Companion flanges (for non-standard sizes) with Fasteners and Gaskets and/or concentric reducers to match process battery limit sizes as per P&ID's. f. Necessary supports for Pump, vessels, exchangers & piping etc. within skid and/or off skid for each package.		Screwed fittings shall not be used in piping for flammable materials
14.	☒	All bolts, gaskets etc. within battery limits and all support and fastening materials to install the various components.		

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Sr. No.		Description	Included in Scope	
			Yes / No / NA	Remarks
15.	☒	Necessary support / cleats for thermal insulation and electrical tracing (if necessary).		
16.	☒	Noise Attenuation means (Acoustic enclosure, insulation etc.) to limit the overall package noise level as project noise requirements.		
17.	☒	Name plate & safety signs (in Kazakh, Russian & English language.		
18.	☒	Safety guard, covers, safety instruction and safety sign where required.		
19.	☒	External surfaces cleaning, preparation and painting with primer, intermediate and final coat, including prefabricated loose parts		
20.	☒	Minimum Inspection and Shop Test as set in this specification		
21.	☒	Preservation for transportation and On-Site storage		
22.	☒	Export packing and marking (seaworthy and/or land-worthy) as specified in the project specification (or as minimum suitable for 12 months storage)		
23.	☒	Design and supply of all necessary provisions for supporting the equipment and components during transport and handling: case, boxes, shipping lugs and bracings		
24.	☒	Lifting lugs / trunnions all including relevant design and certifications, required for handling, loading, unloading and erection of each section, assembled equipment and single lift skid/baseplate		
25.	☒	All special tools and tackles required for installation, testing, commissioning, operation, and maintenance.		
26.	☒	Positive material identification (PMI) as applicable within the package in accordance with referenced Project specification		
27.	☒	VENDOR shall provide only one connection at each package / skid battery limit for each fluid with isolation valves (Full bore ball valves). Any distribution pressure & flow control, Filters PSV are in VENDOR scope.		

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Sr. No.		Description	Included in Scope	
			Yes / No / NA	Remarks
28.	☒	Anti-vibration pads (if any), adjustment screws, liner, pre-cut SS shims of all sizes etc.		
29.	☒	Documentation required for engineering, handling, transportation, storage, erection, commissioning, operation, maintenance and troubleshooting as per **Attachment 2** (as a minimum).		

6.2 SPARE PARTS & SPECIAL TOOLS

VENDOR to supply spare parts applicable to overall scope of supply for the equipment including its accessories, piping, electrical, instrumentation items, etc.

The following applies:

a. Construction / Commissioning spare parts shall be proposed and quoted with separate price and subject to review during bid evaluation; the ones selected will be included in base scope of supply and shall be delivered with the main equipment.

In any case, Vendor shall provide as minimum:

STATIC EQUIPMENT (both standalone process static equipment and static equipment installed in rotating equipment/packages (pressure and non-pressure vessels):

Spare Parts for Erection, Pre-commissioning, Commissioning, Start-up (for each Equipment)	
Description	**Quantity**
A) General for all items	
Gasket made from graflex material, for every type installed	100%
Spiral wound gaskets up to 2 inches class 300 and class 600, for every type installed	100%
B) Shell & Tube Heat Exchangers ("U" type included):	
Gaskets, for every type installed	2 set
Fasteners, bolts and nuts, for every type installed	10% (min. 2. pieces)
Test rings, test flanges and packing for removable tube bundle and floating head type exchangers, if necessary	1 set per each exchanger (Except that for stacked exchangers one set per stack shall be supplied)
C) Pressure Vessels:	
Gaskets, for every type installed	1 set

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Fasteners, bolts and nuts, for every type installed	10% (min. 2 pieces)
D) Filters:	
Filtering elements (disposable type)	2 set (each type)
Gaskets, for every type installed	1 set
Fasteners, bolts and nuts, for every type installed	10% (min. 2 pieces)

PUMPS (Note 1)

- Special tool (if required) for disassembling of impellers and bearings

- 1 set of gaskets, O-rings, sleeves for each item

- 2 sets of mechanical seals for each item

- 2 repair kits of end sealing for each item

- 1 set of glass oil levels on the bearing housing of each item

- 2 sets of Membranes hydraulic accumulator and 10% reserve (or 1 per type as minimum whichever is greater)

- 1 set of bearings (both radial and thrust) for each item

- 1 set of coupling flexible elements for each type/size

Maximum unification of mechanical seals manufacturer is required.

~~**LUBE OIL UNIT**~~ (Note 1)

- 2 spare set of membranes for each item of control valves (if any)

Note 1: *Considering the items' nomenclature as XX-P-XXXX A/B/…, the following shall be noted:*

- *"each item" = 1 set/piece common for both A/B/...*

- *"each pump/compressor/…" = 1 set/piece for item A, 1 set/piece for item B, 1 set/piece for …*

b. Operating spare parts (2 years) shall be quoted as option in accordance with **VENDOR** experience and subject to a separate PO to be finalized during execution according to "Special Purchase Conditions" (LATER).

c. Capital spare parts shall be quoted with separate price as per vendor recommendation and subject to review during bid evaluation; the ones selected included in base scope of supply

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and delivered with the main equipment(s).

Note: Vendor shall provide Metallic Container with N2 bottles and pressure gauge for shafts and impellers

List of all spare parts shall be filled electronically & submitted in the Spare Parts and Interchangeability Record (SPIR) format during detail engineering.

All special tools shall be demonstrated to the Contractor's witness during disassembly/assembly of the equipment. Special tools shall be included in **VENDOR** base scope of supply.

6.3 SERVICES INCLUSION

VENDOR's scope of service shall include as a minimum, but not limited to the following:

a. Attendance to Kick-off Meeting (KOM), Pre-Inspection Meeting (PIM), HAZOP, SIL, 60% and 90% 3D model review meeting, Pre-Painting meeting and any Engineering Meetings as **OWNER** and **CONTRACTOR** deemed necessary to develop and monitor the project during its execution,

   - ✓ Engineering Meeting, as a rule, will be held at **CONTRACTOR**'s office and will be attended by **VENDOR** and **SUB-VENDOR** as deemed necessary based on purpose,
   - ✓ Kick off meeting shall be held at **CONTRACTOR**'s office within 30 days from PO acceptance. **VENDOR'** Project Manager, Engineering and Inspection representative are expected to attend,
   - ✓ PID review meeting shall be held in a single session at **CONTRACTOR**'s office in presence of **OWNER** and **CONTRACTOR** responsible departments.
   - ✓ HAZOP and SIL review shall be held in a single session at **CONTRACTOR**'s office in presence of HAZOP/SIL Chairman, **OWNER** and **CONTRACTOR** responsible departments. Impacts from HAZOP/SIL review outcomes related to safety and environment shall be considered in **VENDOR**'s quotation.
   - ✓ Attendance to the 60% and 90% 3D model review to be held at **CONTRACTOR**'s office in presence of **OWNER** and **CONTRACTOR** responsible departments. **VENDOR** shall also participate during Contractor internal model reviews sessions (60% and 90%) for validation/recommendations.
   - ✓ Pre-painting could be held at painting workshop (to be agreed before Purchase Order issue).

b. Assignment of a dedicated team for the duration of the Purchase Order including as a minimum a Project Manager acting also as dedicated point of contact with **CONTRACTOR,**

c. Development of 3D Model for **CONTRACTOR** approval

d. **VENDOR** to update relevant document in as built revision according to Owner/Site comments received during the equipment erection. **VENDOR** shall submit the native file of relevant document for **CONTRACTOR** reference, without any commercial impact on **CONTRACTOR**.

e. Develop all drawings, documents, calculations, Installation/Operation/Maintenance manual(s), Certification Dossier, Manufacturer Data Book, in accordance and issued for the purpose stated

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at Vendor Required Documentation List (VRDL) included in this this specification and as specified on project procedure 4503-YZ-PC-T0000003 (SLN-2725-0000-00-Z-0003) "VENDOR DOCUMENT CONTROL PROCEDURE".

f. Preparation and submission of package piping isometrics

g. Site erection drawing for the loose supplied items

h. Review and approve any document issued by **SUB-VENDORS** before submission to **CONTRACTOR,**

i. Provide Site Inspection and Test Plan defining the quality activities to be performed at the different stages of erection, pre and commissioning,

j. Provide an Installation specification on **CONTRACTOR**' format, including all information and quantity for the site erection works for the scope of supply covered by the PO, including but not limited to:

- ✓ weight of equipment and single lift skid and baseplate
- ✓ weight of steel structures per grade of fabrication
- ✓ quantity for insulation (unless installed at **VENDOR'** workshop)
- ✓ quantity for tracing (unless installed at **VENDOR'** workshop)
- ✓ quantity for fire proofing
- ✓ quantity for painting at site

k. Preventive maintenance activities table (as part of Maintenance Manual) to detail preventive maintenance activities and including frequency, duration, necessary spare parts.

6.4 EXCLUSIONS

a. All Civil works (foundation and grout)

b. Field Erection / Installation of the Equipment and loose components,

c. Motor Control Centre and electrical power cabling

d. Plant control and emergency shut down systems (DCS/SIS) and MPS/CMS (System-1)

e. Utility supply outside battery limit,

f. Power cabling from electrical sub-station to electric users' terminal boxes

g. Instrumentation interconnecting cabling from remote panels / SIS/DCS/MPS to equipment local junction boxes.

h. Fire detection and protection

i. Foundation Bolts and Nuts unless they are special type not covered by project specification,

j. Material for fire proofing

k. Lubricants and consumables

6.5 FOUNDATION BOLTS

The foundation bolts and nuts are not in the **VENDOR'** scope of supply unless they are special type not covered by project specification; however, the size and the number of anchor bolts shall be verified

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/ calculated by **VENDOR** in accordance with project standard 4503-AA-DX-T0000001 (SLN-2725-0000-10-W-0001 "Standard Anchor Bolts General Information" and 4503-AA-DX-T0000002 (SLN-2725-0000-10-W-0002) "Standard Anchor Bolts Details" and shall be indicated in the first issue of Foundation Drawings. If for any reason, special foundation bolts are required by **VENDOR**, those shall be included in **VENDOR'** scope of supply and delivered in due time for installation at site before equipment delivery.

6.6 LUBRICANTS AND CONSUMABLES

VENDOR shall provide specifications, quantities and change frequency for pre-commissioning, commissioning, start-up, and operational consumables, such as lubricants, chemicals, flushing fluids, fillings, recording chart, etc.

VENDOR shall inform if a dedicate flushing oil must be used for oil flushing system.

Lubricants including oil, grease, hydraulic oil, hot oil, flushing oil shall be listed in the lubricant list per Project format. Electronic form shall be provided by **CONTRACTOR** during detailed engineering.

Lubricant products to be indicated shall be from indicate the brand, unless otherwise approved by **OWNER**.

If any lubricant or consumable is **VENDOR'** exclusive product, **BIDDER** shall inform within the quotation and provide separate quotation for initial charge and price list for the quantity required in operation**.**

6.7 TRAINING

Vendor shall provide training including supply of training material for OWNER Personnel at Vendor workshop / Site. Vendor shall provide basis of the quotation viz. no. of days considered, no. of participants, extent of training coverage etc.

6.8 SEPARATE PRICE

BIDDER is requested to quote the following with separate price:

a. Construction / Commissioning spare parts,
b. Capital spare parts,
c. 2 years spare parts
d. Site assistance on "lump sum basis" as per paragraph 8
e. Site assistance on "daily rates" as per paragraph 8
f. Exclusive product lubricants or consumables,
g. Training
h. Any major test significantly affecting cost and delivery,
i. STP/NWD file of 3D Model as per **VENDOR** Standard format
j. Electrical Heaters, if any, as described in paragraph 4.4.

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7 SUB-VENDOR LIST

BIDDER shall include in the bid a list of proposed **SUB-VENDORS,** also specifying the country of origin**,** for **CONTRACTOR** and **OWNER** review. Only the ones accepted shall be used for the execution of the Purchase Order.

8 SITE ASSISTANCE

VENDOR shall assure its qualified and skillful personnel and/or personnel of its **SUB-VENDOR**(s) for supervision and assistance during erection, pre-commissioning, commissioning, start-up and test run to the PLANT, where necessary.

Both during proposal and as part of official engineering documentation, **VENDOR** shall provide a detailed plan, including relevant time schedule, for the site assistance deemed necessary considering the following:

a. Construction Phase:
   - ✓ Estimation of expected site mobilization (Man/Weeks),
   - ✓ Pre-Requisites for **VENDOR**'s Intervention (i.e. foundations completed),
   - ✓ Plan of activities to be performed by **VENDORS** (i.e. unpacking and check of delivery, check of installation on foundation of main and auxiliary equipment, check of temporary alignment, grouting, etc.);

b. Pre-commissioning Phase:
   - ✓ Estimation of expected site mobilization (Man/Weeks),
   - ✓ Pre-Requisites for **VENDOR**'s Intervention (i.e. availability of utilities, mechanical erection completed),
   - ✓ Plan of activities to be performed by VENDORS (i.e. verification of correctness installation of main and auxiliary equipment, equipment alignment, oil flushing, etc.);

c. Commissioning Phase:
   - ✓ Estimation of expected site mobilization (Man/Weeks),
   - ✓ Pre-Requisites for VENDOR's Intervention (i.e. pre-commissioning activities completed),
   - ✓ Plan of activities to be performed by VENDORS (i.e. auxiliary equipment functional test, nitrogen run, first run, etc.);

Based on what above, VENDOR must consider to:

- Develop detailed procedures for the completion of tasks above at the site ensuring that these procedures indicate the necessary materials, tools, equipment, consumables, utilities and Sub-contractor manpower required to perform the tasks above.
- Evaluate to provide the inspections / technical advisor for activity at site by remote using smart helmet / virtual reality tools / remote assistance.
- Develop a Site Inspection Test Plan
- Provide schedule reporting:
  - all activities in chronological order, referencing each specific task based on the Site Inspection Test Plan (ITP).
  - estimated duration and the number of Vendor Supervisors needed, specifying the skills required for each activity.

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- necessary predecessors/prerequisites for any activity to be carried out by either the CONTRACTOR or VENDOR.

- include a mechanical run-in procedure for rotating equipment/packages using non-dangerous fluids such as water, air, or nitrogen. This procedure is intended to check the mechanical integrity of the equipment before releasing it for operational activities. It is not necessary to verify the performance of the equipment but to confirm its readiness for operation.

The content and requirements of the site assistance plan applicable to the specific scope of supply will be reviewed between CONTRACTOR and BIDDER during the technical evaluation phase. The case where CONTRACTOR does not make use of the **VENDOR**'s site assistance service, in whole or in part, does not constitute ground for **VENDOR**'s loss of the equipment warranties and guarantees of as defined in the P.O.

9 PREFABRICATION AND ASSEMBLY

Each equipment shall be fully assembled on a single transportable skid (or multiple skids in case the single solution is not possible) to ensure that site assembly works is minimized.

Shipment limitations are as follows:

- ✓ Length: 13.6 m;
- ✓ Width: 2.5 m;
- ✓ Height: 3 m;
- ✓ Gross Weight: 21500 kg

BIDDER to confirm limits are not exceeded for each equipment and/or provide alternative configuration in case same is not possible. Any deviation must be declared in the offer and discussed/agreed before the Purchase Order issue.

10 DESIGN GENERAL REQUIREMENTS

Refer to document number 4503-GZ-SG-T0000001 (SLN-2725-0000-56-S-0001) "General Specification - Rotating Equipment and Packages" for the applicable requirements.

10.1 AREA CLASSIFICATION

Electric motor, electrical equipment and materials, instrumentation and ancillaries as well as mechanical equipment shall be as minimum suitable for installation in hazardous area indicated in data sheets attached to Material Requisition. Also, please refer paragraph 1.1 of this specification.

10.2 PIPING

The packaged equipment layout shall meet this specification in terms of required operation and maintenance accessibility and lifting facilities.

Piping shall be routed considering the insulation thickness, equipment accessibility and maintainability requirements.

The tie- in points on packaged equipment shall be brought to the edge of the skid and terminate with flanges.

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Piping installed on skid shall be delivered welded, mounted, tested and painted.

Flanged connections shall be used. Threaded connections are not allowed for process pipe.

Welded connections shall be butt-welded, except on non-hazardous and non-flammable fluids.

Connections at **VENDOR**'s scope limit shall be flanged as per the relevant CONTRACTOR's piping class.

On-skid piping shall be run to the edge of the base plate and shall terminate with a flange.

All nozzles provided on casings, vessels, headers, piping, etc. shall be NPS ¾" minimum.

Non-Standard Sizes like 1¼", 2½", 3½", 5", 7", 9", 22", 26", etc. shall not be used.

All necessary pressure safety valves shall be identified and defined by the **VENDOR**. Unless otherwise specified in the scope of supply, all necessary pressure safety valves shall be included in **VENDOR**'s scope.

For equipment/components weighing more than 20kg, crane access or, alternatively, davits shall be provided.

Piping shall be laid out so that equipment such as control valves, relief valves, start-up strainers, pump casings and shafts, heat exchanger bundles, compressor cartridges/ pistons, etc., can be removed with a minimum amount of dismantling and without providing additional supports.

Platforms shall be provided for valves or equipment which cannot be reached from grade level.

Piping shall be arranged to eliminate overhead obstructions and tripping hazards. Minimum clear headroom for overhead piping shall be 2.2m.

Piping supports, valve stems, valve hand wheels, etc., shall be arranged such that they do not obstruct or interfere with operation and maintenance areas.

Side access ladders are preferred to front access ladders.

All piping shall be adequately supported, considering both operating conditions and transportation/erection. Piping routing and relevant supports shall be designed to ensure thermal expansion and contraction of piping and to avoid sagging, vibration and strain on equipment by means of piping stress analysis, where necessary. Whenever required, it shall be possible to adjust supports so that they may be set correctly.

A sufficient number of permanent supports shall be provided at the following locations to prevent excessive forces and deflection of piping when flanges are disconnected:

- ✓ the inlet and outlet of a safety valve
- ✓ spectacle blind or blind plates and spacers
- ✓ equipment nozzles
- ✓ upstream and downstream of control valves
- ✓ piping components that must be disconnected or replaced during maintenance

VENDOR shall design the piping within the equipment battery limit and shall provide the maximum allowable piping forces and moments, and displacements which may be applied for all battery limit connections. As general rule, the piping will be anchored at skid battery limit.

All connections for auxiliaries (including drain - vent flushing - blanketing) shall be provided at single point supply and single point return for each **VENDOR** skid and located at each baseplate edge. Further distribution & combining within skid, if interconnecting piping is in **VENDOR**'s scope of supply, shall be done by **VENDOR**. In case the interconnecting piping is excluded from VENDOR's scope of supply, **VENDOR** shall endorse **CONTRACTOR**'s interconnecting piping routing.

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10.3 MECHANICAL

- The whole package shall withstand the temperature of minus 38 °C for transportation, storage and erection (lifting) as a rule.

  Vendor shall identify and list the items which do not withstand minus 38 °C for transportation, storage and erection (lifting) (if any). When items are identified, minimum acceptable temperature shall be defined for each of these items.
- U Stamp must be provided for all Pressure Vessels and Exchangers with a design pressure above 3.5 bar g.
- Utilities like Instrument Air, Plant Air and Nitrogen will be available at package battery limit with a temperature which could reach a minimum of -38°C depending on seasons. Vendor shall identify and list the items which do not withstand this minimum temperature. When items are identified, minimum acceptable temperature shall be defined for each of these items.

  Vendor is required to verify if a suitable technical solution can be provided (for instance the installation of electric heaters with temperature control start/stop on each utility inlet nozzle or at the inlet of the whole package) and quote it as an option.
- All Lube Oil Piping with flanges, valves, sight glasses, nuts and bolts required in SS316L material.
- For both process and auxiliaries gas/fluids, double safety valves, whit key interlocking system (on both inlet and outlet valves), shall be installed except for the items provided with installed spare. For those items, single safety valve is acceptable.
- All equipment must be provided with valve and blinded vent & drain connections (equipment's such as heat exchangers, filters etc.).

10.4 ELECTRICAL

Electrical components shall conform to the specifications listed in the Supply Specification and shall be suitable for the specified hazardous area classification and protection degree.

The maximum power requirement of the driven equipment shall be determined at the driver coupling and shall include all transmission losses.

Driver and transmission shall be sized for end of curve operation in case of end of curve operation is specified in the data sheet or were deemed necessary by the Vendor.

VENDOR shall ensure earthing continuity of packaged equipment.

Earthing lug shall be provided by the **VENDOR** according to CONTRACTOR's standard.

The scope of supply must include:

a. Electric Motors.

b. Soft starter mandatory for:

   - LV motors rated from 150 kW and less than 400 kW.

c. Cable glands with necessary reducers / enlargers as per final cable sizes only for cables supplied by Vendor. With reference to cables provided by CONTRACTOR, the cable glands will be supplied by the CONTRACTOR.

d. Instrumentation (PTCs /RTDs for windings, BTDs, MMS probes) for main motor.

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e. Electric Continuity and Earthing Lugs

f. Earthing provisions for all components – minimum 2 nos. earth bars on each skid/equipment at diagonally opposite positions.

g. Electrical space heater for motors, as applicable.

h. Field Junction Boxes (including Transition junction boxes, if required, for armored cables), wiring up to junction boxes.

i. Transformers and/or other auxiliary or power devices necessary to generate voltages different from the supplied ones, in particular.

j. All assembly cabling and wiring within the package battery limits, including also cable trays for routing of Owner/Contractor power supply feeder and control cables.

k. Electrical heat tracing for Pump and interconnecting piping with necessary supports, fully insulated and cladded with single point electrical termination.

l. Equipment, as applicable, as defined in Cl. No. 12 (Electrical Battery Limit) of Specification 4503-NM-SG-T0000001 (SLN-2725-0000-65-S-0028).

m. Partial Discharge Detectors/system for all Medium Voltage Induction and Synchronous Motors Monitoring of the partial discharge shall be carried out in "On-line" mode for direct starting electrical motors (DOL) and in the "Off-line" mode for electrical motors equipped with Variable Speed Drive or Soft Starter.

10.5 INSTRUMENTATION AND CONTROL SYSTEM

Instrumentation and control system shall conform to Instrumentation and Automation specifications listed in the Supply Specification.

Instrumentation and Automation deliverables required as per current Supply Specification shall be in accordance with the attached samples (LATER).

All necessary instrumentation for proper functioning of the equipment under all conditions and for the execution of performance tests shall be supplied by the **VENDOR** and suitable for the specified hazardous area classification.

Supply of instrument shall include cables up to junction boxes, junction boxes, process tubing, pneumatic tubing, heating boxes, sunshade, channels from instrument till junction boxes or manifold Battery Limit, tagging, winterization, isolating valves, etc. Unless otherwise specified, the local instruments shall be wired to equipment mounted junction boxes.

VENDOR Junction Boxes / Local Panel will be connected to DCS / SIS systems by means of Smart Junction Boxes supplied by CONTRACTOR and located in field. VENDOR SHALL engineer and supply package Junction Boxes with proper dimensions and quantity cable entries to permit CONTRACTOR to connect all signals from Package Junction Boxes to Smart Junction Boxes by means of single pair cables. Cables outgoing from Junction Box shall be single pairs, Junction boxes shall have suitable number of cable entries to permit these connections.

Multicore cables will not be used for connecting Package Junction Boxes to Smart Junction Boxes.

All field instrumentation shall be securely mounted so that they are not affected by vibration, are properly visible during normal operation and are easily accessible for maintenance.

Wherever heat tracing of impulse line is required, according to Winterization Philosophy, it shall be done by means of electrically traced tubing bundle (special factory-made cable in which pulse line is

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installed together with heat tracer in the layer of insulation material with outside polymer protective coating).

Instrument mounting/hook-up shall allow instrument venting and drainage for calibration and isolation for replacement while the machine is in operation.

The scope of supply shall include:

a. Local control panel for startup / maintenance / operation of complete package.

b. All Instrumentation needed for monitoring of process parameters (like PI, TI, FI, SG etc.), machine safety and interlocks (like PT, TT, FT, ST, etc.), completely wired up to Junction Box located at the skid edge supplied by the Vendor.

c. Vibration/Temperature/Axial position probes with Proximeters / transmitters / transducers connected to junction box for API 610 type medium duty process centrifugal Pump package wherever necessary for safe operation and recommended by Vendor as per API 670, Datasheet & Project specs. VENDOR shall provide probes to be connected to Machine Protection System (MPS), Inclusive of wiring within skid and JB at skid edge.
   Manufacturer & Type: MPS/CMS Manufacturer or equivalent, compatible type.

   MPS is supplied by TCM. Vendor to provide information for configuration of CMS.

d. Winterization Including heating box for transmitter and bundle tubing for instrument process connection.

e. Wiring to junction box within machine & auxiliary units. Interconnecting wiring (cables trays, cables, cable glands) from instruments to junction box (JB) within the skid/baseplate limit. Cable glands for **CONTRCATOR's** multicables shall be induced in **VENDOR's** scope. However, VENDOR must foresee cable entries for multicore cables based on information made available by CONTRACTOR. VENDOR is allowed to drill the JB's only after having received the required diameter.

f. Instrument air piping inside of machine and units.

g. Instrument air distribution manifold and instrument air tubing to air consumers in SS metallurgy.

h. Regulator and Filter for each user of instrument air.

i. Interconnecting wiring between instruments and junction boxes.

j. Junction boxes to terminate the instrument wiring.

k. Local gauge board(s), cables, cable trays, junction boxes, bulk materials (cable glands etc) and other instrumentation for complete scope of supply (within package battery limits).

l. All field instruments and valve positioners shall be SMART type transmitters HART protocol (version 7 at least).

m. SIL certificates, SIL certification reports, Safety Manuals for any item involved in a SIF SIL classified (initial elements, logic solver and final elements) and Process Safety Time for any SIF SIL classified belonging to the package in VENDOR scope. These documents shall report failure rates (λSD, λSU, λDD, λDU), diagnostic coverage, MTBF, MTTR, etc.

n. All instrumentation required for safe, trouble free and proper operation, control, protection.

o. Instrument's primary and secondary hook ups.

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p. Loose supplied instrument, item shall be designed taken into consideration the location where it is installed and in compliance with required connection type.

10.6 STRUCTURAL

When the equipment shall be delivered mounted on one or several base plates, they shall comply with the following requirements:

a. Unless otherwise specified, base plates shall be constructed of welded structural steel. With **CONTRACTOR**'s approval, high strength bolts may be used when welding is impractical;
b. Base plates shall be equipped with lifting lugs or devices suitable for a four (4) point lift and shall be of sufficient strength and rigidity for transportation and installation to avoid any distortion or damage to the equipment or internal parts of machinery installed on them during lifting;
c. The base plate shall be designed to withstand both static and dynamic stresses of machinery and to transmit equipment generated forces and couples to foundation. Equipment mounted on the base plate shall not be considered as contributing to its bracing.
d. Base plate shall be equipped with leveling screws for installation on concrete foundation and alignment screws and shims for equipment alignment.
e. Unless otherwise specified, base plates shall be designed for grouting and shall have the necessary lipped grouting hole per each bulkhead section and appropriately located vents. The base plate bottom shall be coated with a suitable coating system compatible for epoxy grout irrespective of different grouting material followed below skid.
f. The projection of all the equipment included in the skid shall fall within the maximum perimeter of the base plate.
g. Structures shall be designed to permit approach to process equipment and to lift it without obstruction.
h. Whenever necessary the skid structure will include the support beams, to support the external piping. These will be properly designed and installed as per **CONTRACTOR** comments and mark-ups.

11 NOISE

Unless specified otherwise in applicable specification, the guaranteed noise levels and noise control design, for all running conditions, shall be in accordance with specification 4503-SZ-SG-T0000004 (SLN-2725-0000-30-S-0001) "Noise General Specification". **VENDOR** shall indicate guaranteed sound power and pressure levels for different octave bands as well as overall A- weighted values, to insert in the Noise Source Data sheet format attached in 4503-SZ-SG-T0000004 (SLN-2725-0000-30-S-0001) "Noise General Specification".

If a part of the supply does not meet the job noise limit, **VENDOR** shall provide the necessary noise attenuation devices (acoustic hoods, acoustic insulation, silencers, etc.) **VENDOR** shall be responsible for the selection of equipment to meet the requirements including modification of design, acoustic insulation and provision of enclosures.

In case acoustic hood is required for noise attenuation, **VENDOR** shall ensure that all battery limit connections are located outside the enclosure. Local instruments shall also protrude outside the acoustic enclosure to facilitate reading and replacement. Foundation inputs for design (by **VENDOR**) must also take into account the provision for installation of acoustic hood. **VENDOR** shall submit

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dimensional drawing of acoustic hood showing its fixing details on the foundation. Ventilation fans 2 nos. (1 working + 1 standby) for noise hood, adequate lighting, gas detector with visual and sound (only in case of equipment handling nitrogen or hydrocarbon) shall be provided by **VENDOR**.

12 PAINTING

Unless otherwise specified, all equipment and material, Loose components, electric motors, electric equipment, control panels, instruments etc. shall be shipped fully protected and painted up to finish coat in accordance with specification 4503-VW-SG-T0000001 (SLN-2725-0000-25-S-0001), "General Painting Specification".

The extension of painting shall be maximized up to the finish coat for all the prefabricated equipment (for example skid mounted packages, pressure vessels, machinery, prefabricated piping, supports, etc.), that will require only a final assembly and/or connection at battery limits in the field.

Only bulk material that will be delivered loose at site and which requires site welding, shall be delivered painted with final coat keeping unpainted spaces for welding area.

Fabricated components shall not be painted until the inspection of the parts is completed**.**

13 INSULATION

VENDOR shall identify on its PID's and General Arrangement Drawings the parts to be insulated / heat traced, even for parts to be insulated / heat traced by others.

Parts to be insulated shall be designed for the installation of insulation (provision of insulation clips, over length of instrument nozzles and valve hand wheels etc.) in accordance with Specification 4503-XL-SG-T0000001 (SLN-2725-0000-55-S-0004) "Insulation General Specification".

After the Purchase Order, the **VENDOR** shall provide the total length and diameters of lines and dimensions of equipment to be insulated and/or heat traced.

The **VENDOR** shall ensure to provide adequate space around the piping / equipment to be insulated at site by **CONTRACTOR**, so that dismantling of any component is not required for application of insulation.

Monitoring instruments like pressure gauges, level indicators etc. (switches and transmitters excluded) will be insulated in such a way that instrument reading and access for maintenance are not hindered.

Flanged connections and rotating parts (couplings, seal, stuffing boxes etc.) shall not be insulated unless specifically required for process reasons, they shall be provided with flanges covers. Valves shall be insulated with valve caps for easy removal.

Insulation shall be applied only after completion of testing.

VENDOR shall provide, for equipment insulation, a typical drawing and layout where the exact supports position for the external finishing and insulation is indicated. The supports design shall guarantee the correct site installation of the insulation system, avoiding any field additions/modifications.

For piping and equipment insulated for personnel protection with metal barrier, the supports shall guarantee the strength of the protective structure once it is installed.

14 3D MODEL REQUIREMENTS

In the base scope, **VENDOR** shall submit 3D model of the supplied equipment and piping as per

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VENDOR standard format / *.STP file.

As optional scope, **VENDOR** shall provide a complete project review file in accordance with one of the following electronic graphic format and in fully compatible with the Plant coordinates:

- ✓ *.ifc,
- ✓ *.sat (with data)
- ✓ *.vue + *.mbd2/xml
- ✓ *.rvm + *.att
- ✓ *.dwfx (with data)
- ✓ *.zvf

All connecting points at battery limit (i.e., equipment nozzles, inlet and outlet flange of skid mounted packages etc.) shall be modeled and clearly identified. All access and maintenance clearance shall be shown in the 3D model.

For skid mounted packages, an outline shape of the package is enough, provided that the connecting points are clearly defined.

VENDOR to consider following information applicable for 3D model:

- ✓ Skid mounted equipment to be modelled with exact geometry
- ✓ Skid to be tagged as main equipment.
- ✓ All sub-equipment of all skids to have skid tag as a prefix.
- ✓ All sub-equipment to be modelled with exact geometry.
- ✓ Complete internal Piping of the skid with all inline and online instruments to be modelled.
- ✓ All pipe supports with the skids are to be modelled

Model shall be submitted progressively 2 weeks before the 3D model review sessions (30%, 60%, 90% and 100%) as per **CONTRACTOR** project schedule.

The 3D model provided by **VENDOR** shall contain, as a minimum, the following attributes:

- ✓ tag of the equipment and its subcomponents
- ✓ Description of the equipment
- ✓ Geographical location of the eq

Description	3D Model Category
API 610 Type Medium duty Process Centrifugal pumps	Category 3

15 GUARANTEES / WARRANTEES

The equipment shall be guaranteed for all the performance parameters listed in the respective datasheets. Performance parameters to be matched with the approved final equipment datasheet. Overall package power and utilities consumptions shall be guaranteed by **VENDOR** and will be subject to the following rule:

Service	VENDOR's Guaranteed Value	Liquidated Damages Type (to be defined and included in the Purchase Conditions)
Overall Cooling Water (m3/h)	To be defined by **VENDOR**	With threshold up to 2.5%, **VENDOR** shall start to pay LD at 2.5% above the guaranteed value and for each full 2.5%. LD amount to be discussed and agreed before Purchase Order issue.
		Exceeding the guaranteed value above 10% for each guaranteed product specific energy consumption, the API 610 Type Medium duty process centrifugal Pump Package will be subject to "make good".

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Service	VENDOR's Guaranteed Value	Liquidated Damages Type (to be defined and included in the Purchase Conditions)
Overall Electrical Absorbed Power (kW)	To be defined by **VENDOR**	With threshold up to 2.5%, **VENDOR** shall start to pay LD at 2.5% above the guaranteed value and for each full 2.5%. LD amount to be discussed and agreed before Purchase Order issue.
		Exceeding the guaranteed value above 10% for each guaranteed product specific energy consumption, the API 610 Type Medium duty process centrifugal Pump Package will be subject to "make good".

VENDOR shall demonstrate the capability of the Package to fulfil the guaranteed process performances and the utility consumption rates as per Mechanical Data Sheet/Duty Specification during a test run of 72 hours to be performed in field after the Plant has been successfully started up and steadily operated.

VENDOR shall submit a test run procedure for approval before delivery. Additional instrumentation necessary for the test run are in **VENDOR** scope.

All equipment and components including electrical, instrumentation and structure shall be guaranteed against design and manufacturing defects. Manufacturer shall give free replacement of any part/component that is found to be faulty during the above referred guarantee period. Such replaced parts shall also be subject to a guarantee for a minimum period as indicated in LOI/PO.

VENDOR shall ensure that all equipment, including that furnished by sub-supplier, is furnished to the applicable specifications referenced on the datasheet or material requisition.

16 EXPORT CONTROL AND INTERNATIONAL SANCTIONS

As soon as possible the VENDOR shall submit the draft Export Compliance Details (as per the EXPORT COMPLIANCE DETAILS FORM attached to the Special Purchase Conditions) for review by CONTRACTOR's Export Compliance Officer.

Not later than one hundred eighty (180) days before contractual delivery date, VENDOR shall submit to CONTRACTOR, the Export Compliance Details as reviewed by CONTRACTOR's Export Compliance Officer.

The EXPORT COMPLIANCE DETAILS FORM shall detail the scope of supply at least as specified below:

- In case of bulk materials or tagged items, as per the positions of the Purchase Order,
- In case of packages, detailing each item shipped loose or detachable,
- In case of spare parts and U.S. components, detailing each item.

17 INSPECTION AND TESTING

17.1 GENERAL

Inspection and testing shall be carried out in accordance with the accepted Inspection & Test Plan.

VENDOR is responsible for inspection, testing activities and certification also of his **SUB-VENDORS**.

At any time during the fabrication and testing phase, **CONTRACTOR/OWNER** or his designated representative shall have the access to **VENDOR'**s and any **SUB-VENDORS** workshop.

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17.2 ITP (INSPECTION TESTING PLAN)

CONTRACTOR provides to **VENDOR** the ITP Standard (Inspection requirements in compliance with **CONTRACTOR** standard practice). Refer to Attachment 1.

The ITP Standard requires to be checked, filled-in and implemented by **VENDOR** in accordance with the contractual specifications and inspection requirements.

VENDOR shall resubmit it to **CONTRACTOR** along with the technical offer.

The following shall be considered:

a. Any changes in ITP Standard format are not allowed;
b. ITP Content and requirements applicable to the specific scope of supply will be checked between **CONTRACTOR** Inspection Department and **VENDOR** (at presence of the QC key personnel) during the technical evaluation phase;
c. After the Purchase Order is formalized, **VENDOR** shall submit the ITP to **CONTRACTOR** for review & acceptance within the contractual terms as per Supply Specification (Vendor Required Document List);
d. The ITP acceptance does not relieve in any case **VENDOR** from his responsibilities to comply with all tests, inspections and certifications requirements defined in the applicable job specifications and codes;
e. Regardless involvements foreseen in the accepted ITP, **CONTRACTOR** or his appointed representative, **OWNER** and Third Party (if any) reserve the right to witness inspection at any stage of fabrication;
f. **CONTRACTOR/OWNER** review does not constitute acceptance of a deviation unless **VENDOR** have specifically identified the proposed deviation request that shall be submitted to **CONTRACTOR** for acceptance;
g. **VENDOR** shall consider **CONTRACTOR/OWNER** involvements foreseen in the ITP Standard as a minimum requirement, however **CONTRACTOR/OWNER** can provide improvement of the HOLD, WITNESS and REVIEW points without any additional cost and extension of lead time;
h. **OWNER** has the right to employ a Third Part Inspection Agency for Inspection. The inspection by Third Party Inspection Agency, when required, shall be at **OWNER**'s cost and expenses;
i. Inspections and tests, when witnessed by **CONTRACTOR** and/or **OWNER** and/or their appointed AGENCIES, shall not relieve **VENDOR** from his responsibility to ensure the compliance of quality and workmanship of the equipment/material supplied to the contractual requirements;
j. **VENDOR** is responsible of its sub-supplies, in terms of coordination and surveillance of inspection activities. **VENDOR** shall arrange a suitable inspection and testing plan for all bought items from their **SUB-VENDOR**(s). This ITP(s) shall be included/attached in the "main" ITP submitted to the **CONTRACTOR** for review and acceptance (after the Purchase Order).
k. In case, after PO award, test not previously included on ITP but necessary to comply with Laws and/or applicable codes/standards shall be added, **VENDOR** shall consider it already included in the scope of supply and will not give right to any extension of time nor to cost impact.
l. If, during intermediate or final inspection, the test result does not completely fulfil the requirements indicated in the test procedure, **CONTRACTOR** have the right to

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request/evaluate additional inspections or examinations to ensure that equipment complies with the specifications without claiming right to any extension of time nor to cost impact.

17.3 NDE PROCEDURE

NDE procedures and personnel qualifications shall be subject to **CONTRACTOR**'s review/approval and shall form an integral part of the manufacturers QC documentation.

All personnel performing Non Destructive Tests (N.D.T) must be adequately qualified and Record shall be available at **VENDOR**'s shop(s) during the test for Inspector review.

17.4 START OF FABRICATION

The approval status of the ITP does not prevent the starting of the manufacturing activities of the **VENDOR**. It is **VENDOR** responsibility to comply with all tests, inspections and certifications requirements defined in the applicable job specifications and codes.

17.5 PRE-INSPECTION MEETING

Pre-inspection meeting shall be held at Manufacturer Workshop (or as deemed appropriate) only after the ITP is accepted as final by **CONTRACTOR** or accepted with minor comments as a minimum.

17.6 INSPECTION NOTIFICATION

Unless otherwise specified, the **VENDOR** shall notify the **CONTRACTOR** of the expected dates for witness (W) & hold (H) points ten (10) working days in advance.

Notification shall be sent through official email using Project Form for all the activities identified with an inspection point (W/M/H) on ITP for Production.

Monthly Inspection schedule shall be issued by the **VENDOR** along with the Monthly Progress Report.

17.7 TESTING

VENDOR shall perform all tests and inspection necessary to ensure the compliance of material and workmanship conform to the Project Requirements.

The Final Tests (e.g. Hydrostatic, pneumatic, water filling, Complete unit running test (where applicable), F.A.T., etc.) can be performed only if:

a. **VENDOR**'s engineering, welding documents have been accepted by **CONTRACTOR** with no pending technical queries or deviations/concessions;
b. All the NCR have been closed;
c. ITP (for notified item) and relevant test procedure are in final acceptance status;
d. MTC (material test certificates) and test reports have been reviewed/signed by **CONTRACTOR** assigned Inspector.
e. All ITP phases have been duly signed off by **VENDOR**, by the **CONTRACTOR** and/or OWNER and/or THIRD PART AGENCY.

All pressure-containing materials require Positive Material Identification (PMI) .

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17.8 CERTIFICATION DOSSIER

VENDOR shall provide preliminary Index of Certification Dossier (CED) within 4 weeks from P.O. or at KOM whichever is the earliest. Index will be finalized during the PIM to permit collection of certification documents and testing inspection report as per the same on real time basis.

CED book shall be filled in continuously during the fabrication with the aim to have the book ready at the final Inspection. Status of the CED shall be indicated by the **VENDOR** on the Monthly Progress Report and will be verify by the Inspector/Expediter during the Inspection/Expediting visit.

Preliminary copy of CED could be required by **CONTRACTOR** to ascertain that preparation of the book is ongoing.

17.9 IRC (INSPECTION RELEASE CERTIFICATE)

At the completion and acceptance by **CONTRACTOR**'s INSPECTOR of the fabrication, testing and packing of all equipment, **CONTRACTOR**'s Post Order Manager will issue the IRC (Inspection Release Certificate) to authorize **VENDOR** to ship the goods.

18 PROTECTION AND PRESERVATION

18.1 PREPARATION FOR SHIPMENT

It is **VENDOR** 's responsibility to pack the Supply according to the requirements listed in the "Special Purchase Conditions" and to properly protect the Supply from any damage and corrosion considering also the requirements for sea transportation, truck transportation, railway transportation, different handlings and transshipments, transit period and outdoor storage period at Facility site at ambient temperature ranging from -38 ºC and +45 ºC.

The packing shall be seaworthy and suitable for loading by crane, as well as for shipment in freight transport, trailers, and trucks and for the transshipment by forklift, in so far as the weight and volume of individual packages allow.

All internal parts other than stainless steel, shall be coated with a rust preservative for outdoor storage.

All openings, flanged or threaded, shall be sealed airtight prior to shipment. Flanges shall be covered with gasketed wooden blanks, secured with a minimum of four bolts. Plastic plugs are not acceptable.

Exposed machined surfaces shall be coated with a thick film of rust preservative and wrapped where practical with weather-proof tape.

All piping and tubing connections shall be tagged or marked for identification in the field.

All the bearing housings shall be filled with oil or grease prior to shipment. Where it is not feasible, **VENDOR** shall indicate which bearings will have to be lubricated at site.

Shipping box shall have a removable corrosion protection inspection port.

VENDOR shall submit his standard preservation and packaging procedure for **CONTRACTOR** review as per the document's submittal schedule, and at least three months before shipment.

All the documents including the **VENDOR** Data Book shall be in final issue at least one month before the final inspection.

No material or equipment shall be shipped until a release note for shipment has been issued by the **CONTRACTOR**.

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18.2 ON SITE STORAGE

VENDOR shall provide the **CONTRACTOR** with all the instructions required to insure a correct protection of the equipment during on-site storage. These instructions shall include at least:

a. The required site storage conditions following conventional symbols for each package as specified in "Special Purchase Conditions" (for example A= storage in open air, B= storage under shed, C= Storage in closed dry unheated premises, D= Storage in closed heated premises, E= Storage in closed air-conditioned premises).

b. The measures to be taken at site arrival like nitrogen blanketing, oil filling, electrical motors space heater to be energized etc.

c. The periodical inspections/ actions to be performed like rotation of machinery shafts, change of desiccants etc.

d. The preservation requirements once the equipment has been installed but is not running.

19 VARIATIONS OF ORDER

Modifications in the scope of supply, if any, shall be managed according to the Purchase Conditions indicated in the Purchase Order.

After the issue of Purchase Order, any engineering activity concerning the definition or confirmation of all items under "HOLD"/"To be confirmed/to be provided after Order" and the implementation and drafting of **CONTRACTOR/OWNER** comments are not recognized as additional activities.

20 DOCUMENTATION DURING SUPPLY EXECUTION

VENDOR shall supply as minimum all the documentation listed under the Attachment 2. **VENDOR** is responsible to supply/submit the required documentation in accordance with the agreed schedule as per Attachment 2.

Documentation shall be in accordance with the Project specification number 4503-YZ-PC-T0000003 (SLN-2725-0000-00-Z-0003) "VENDOR DOCUMENT CONTROL PROCEDURE".

The **CONTRACTOR** will return the submitted documentation for review/approval within fifteen (15) working days from the receipt.

All documentation returned to the **VENDOR** with comments shall be resubmitted to the **CONTRACTOR** within ten (10) working days from the receipt.

The **CONTRACTOR** review /approval do not relieve the **VENDOR** from his responsibilities to comply with the contractual obligation.

For the Final documentation, **VENDOR** shall follow the Project specification number 4503-YZ-PC-T0000003 (SLN-2725-0000-00-Z-0003) "VENDOR DOCUMENT CONTROL PROCEDURE" for the detail criteria/guidelines for arranging the documentation inside the Final Data Books. Four (4) soft copies (CD) are required for final data books (TO BE CONFIRMED).

20.1 FORMATS TO BE USED

Following project formats shall be used by **VENDOR** during execution for issuing the related documentation:

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- ✓ Clarifications and Deviations format;
- ✓ Vendor documents register format;
- ✓ Monthly Progress Report format;
- ✓ Transmittal format;
- ✓ Noise Source Data sheet format;
- ✓ LV/MV Motor Data sheet format;
- ✓ Soft Starter Data sheet format;
- ✓ Utility Consumption List Format;
- ✓ Electrical Consumption list format;
- ✓ In-tools ISF format for instruments data sheet;
- ✓ Package I&C Database format;
- ✓ Lubricant List format;
- ✓ Inspection and Test Plan (ITP) format;
- ✓ Inspection Notification format;
- ✓ Spare Parts & Interchangeability Record format;
- ✓ Special Tools List format;
- ✓ Document templates of all sizes;
- ✓ PID with project symbology / legends format;
- ✓ Cause & effect charts format;
- ✓ Data Information for Erection at Site format

20.2 TECHNICAL REFERENCE CONTACT

For any technical subject reference shall be made to:

TECNIMONT S.p.A.
Via Gaetano De Castillia 6/A - 20124 Milan, Italy
To the attention of Mr. Marco Lazzari
E-Mail Address: m.lazzari@tecnimont.it
Tel.+39.02.6313.7016

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ATTACHMENT 1 - REFERENCE DOCUMENTS

The documents listed below, and the Specifications, Codes and Standards referenced in these documents (even though not specifically listed here below) are applicable in their entirety unless agreed deviations are listed in deviation list. It is **VENDOR**'s responsibility to ask the **CONTRACTOR** for any document not received within the requisition but referenced inside the below-listed documents.

Pos.	Document Number	Document Title	Issue
		GENERAL	
1.	SLN-2725-0000-31-S-0001 4503-SZ-SG-T0000007	HSE Protection Philosophy	1
2.	SLN-2725-0000-25-S-0001 4503-VW-SG-T0000001	General Painting Specification	A
3.	SLN-2725-0000-30-S-0001 4503-SZ-SG-T0000004	Noise General Specification	0
4.	Refer to the Special Purchase Conditions	Packing Specification	--
5.	SLN-2725-0000-00-A-0001 4503-YZ-SG-T0000001	Basic Engineering Design Data	0
6.	SLN-2725-0000-00-A-0002 4503-YZ-SG-T0000002	Utility and Auxiliary Materials Conditions and Specification	0
7.	SLN-2725-0000-55-Z-0001 4503-FZ-SG-T0000001	Vendor Preservation General Specification	0
8.	SLN-2725-0000-00-Z-0001 4503-YZ-PC-T0000001	Document Numbering Procedure	A
9.	SLN-2725-0000-00-Z-0003 4503-YZ-PC-T0000003	Vendor Document Control Procedure	A
10.	SLN-2725-0000-96-S-0001 4503-IX-SG-T0000001	General Specification for 3D Model	B
11.	SLN-2725-0000-05-Z-0006 4503-ZZ-PC-T0000006	Spare Parts Procedure	A
12.	SLN-2725-0000-00-Z-0004 4503-YZ-PC-T0000004	Tagging Procedure	2
13.	--	Sub-Vendor list for Electrical and Instrumentation Parts	--
14.	SLN-2725-0000-05-Z-0001 4503-ZZ-SG-T0000001	Instruction To Vendors for Obligatory Certification, Permits and Authority Documentation	A
15.	Included in 4503-YZ-PC-T0000003	Vendor documents register format	--
16.	Included in Special Purchase Conditions	Monthly Progress Report format	--
17.	Format	Transmittal format	--

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Pos.	Document Number	Document Title	Issue
18.	Included in 4503-SZ-SG-T0000004 Noise General Specification	Noise Source Data sheet format	--
19.	Format	Utility Consumption List Format	--
20.	Format	Lubricant List Format	--
21.	Format	Inspection and Test Plan (ITP) format	--
22.	Format	Inspection Notification format	--
23.	Format	Spare Parts and Interchangeability Records (SPIR) Format	--
24.	Format	Special Tools List Format	--
25.	Format	Data Info for Erection at Site Format	--
26.	Format	Document templates of all sizes	--
27.	Format	Clarifications and Deviations format	--
MACHINERY / PROCESS			
28.	SLN-2725-5110-56-D-0061 4503-GG-SE-T51P1003	Mechanical Data Sheet for Pumparound Pump Item: 51-P-1003 A/B/C/D	A
29.	SLN-2725-5110-56-D-0062 4503-GG-SE-T51P1004	Mechanical Data Sheet for Passivation Pump Item: 51-P-1004 A/S	A
30.	SLN-2725-5110-56-D-0063 4503-GG-SE-T51P1006	Mechanical Data Sheet for Reactor Effluent Pump Item: 51-P-1006 A/B	A
31.	SLN-2725-5120-56-D-0064 4503-GG-SE-T51P2002	Mechanical Data Sheet for Recycle Column Reflux Pump Item: 51-P-2002 A/B	0
32.	SLN-2725-5120-56-D-0065 4503-GG-SE-T51P2004	Mechanical Data Sheet for Butene-1 Column Reflux Pump Item: 51-P-2004 A/B	0
33.	SLN-2725-5120-56-D-0066 4503-GG-SE-T51P2006	Mechanical Data Sheet for C5+ Product Pump Item: 51-P-2006A/B	0
34.	SLN-2725-5120-56-D-0067 4503-GG-SE-T51P2007	Mechanical Data Sheet for Butene-1 Product Pump Item: 51-P-2007A/B/C	0
35.	SLN-2725-5670-56-D-0068 4503-GG-SE-T56P7001	Mechanical Data Sheet for HP Flare Knock-Out Drum Pump Item 56-P-7001 A/B	A
36.	--	ITP for API 610 CENTRIFUGAL PUMPS	--
37.	SLN-2725-0000-56-S-0001 4503-GZ-SG-T0000001	General Specification - Rotating Equipment and Packages	0
38.	SLN-2725-5110-25-F-2010 4503-VZ-DM-T51102010	Piping and Instrumentation Diagram- LAO - Reaction and Catalyst Removal - Pumparound Loop A	A

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Pos.	Document Number	Document Title	Issue
39.	SLN-2725-5110-25-F-2011 4503-VZ-DM-T51102011	Piping and Instrumentation Diagram- LAO - Reaction and Catalyst Removal - Pumparound Loop B	A
40.	SLN-2725-5110-25-F-2012 4503-VZ-DM-T51102012	Piping and Instrumentation Diagram- LAO - Reaction and Catalyst Removal - Pumparound Loop C	A
41.	SLN-2725-5110-25-F-2013 4503-VZ-DM-T51102013	Piping and Instrumentation Diagram- LAO - Reaction and Catalyst Removal - Pumparound Loop D	A
42.	SLN-2725-5110-25-F-2021 4503-VZ-DM-T51102021	Piping and Instrumentation Diagram- LAO - Reaction and Catalyst Removal - Passivation	A
43.	SLN-2725-5110-25-F-2015 4503-VZ-DM-T51102015	Piping and Instrumentation Diagram- LAO - Reaction and Catalyst Removal - Reactor Effluent Pumps	A
44.	SLN-2725-5120-25-F-2009 4503-VZ-DM-T51202009	Piping and Instrumentation Diagram- LAO - Separation - Recycle Column Reflux Drum Pumps	A
45.	SLN-2725-5120-25-F-2005 4503-VZ-DM-T51202005	Piping and Instrumentation Diagram- LAO - Separation - Butene-1 Column Reflux Drum	A
46.	SLN-2725-5120-25-F-2008 4503-VZ-DM-T51202008	Piping and Instrumentation Diagram- LAO - Separation - C5+ Product	A
47.	SLN-2725-5120-25-F-2007 4503-VZ-DM-T51202007	Piping and Instrumentation Diagram- LAO - Separation - Butene-1 Product Pumps	A
48.	SLN-2725-5670-25-F-2001 4503-VZ-DM-T56702001	Piping and Instrumentation Diagram- LAO - Blowdown and Flare System - High Pressure Flare Vent System	A
49.	SLN-2725-0000-25-S-0030 4503-VZ-SG-T0000003	Process General Philosophies - Winterization Philosophy	A
50.	SLN-2725-0000-25-S-0002 4503-VZ-SG-T0000002	Process Design Criteria and Philosophy	A
51.	--	P&ID Legends and Symbols	--
STATIC EQUIPMENT			
52.	SLN-2725-0000-55-S-0001 4503-CZ-SG-T0000001	General Specification - Static Equipment	1
53.	SLN-2725-0000-55-S-0002 4503-CZ-SG-T0000002	General Specification - Static Equipment Nameplate	0
CIVIL AND STRUCTURAL			
54.	SLN-2725-0000-10-W-0001 4503-AA-DX-T0000001	Standard Anchor Bolts General Information	A
55.	SLN-2725-0000-10-W-0002 4503-AA-DX-T0000002	Standard Anchor Bolts Details	A

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Pos.	Document Number	Document Title	Issue
ELECTRICAL			
56.	SLN-2725-0000-65-A-0001 4503-NN-SG-T0000001	General Design Criteria for Electrical Plant	A
57.	SLN-2725-0000-65-S-0026 4503-NN-SG-T0000003	General Specification - Selection Criteria for Electrical Equipment Installed in Hazardous Area	A
58.	SLN-2725-0000-65-S-0020 4503-NM-SG-T0000002	General Specification - LV Induction Motors	A
59.	SLN-2725-0000-65-S-0028 4503-NM-SG-T0000001	General Specification – Electrical Requirements for Mechanical Packages	A
60.	SLN-2725-0000-65-D-0025 4503-NM-SE-201T01001	Data Sheet - LV Motor	A
61.	SLN-2725-0000-65-S-0017 4503-NK-SG-T0000003	General Specification - LV Cables	A
62.	SLN-2725-0000-65-S-0025 4503-NM-SG-T0000007	General Specification – Soft Starter	A
63.	SLN-2725-0000-65-D-0039 4503-NM-SE-211T01001	Data Sheet – Soft Starter	A
64.	--	Electrical consumption list format	--
INSTRUMENTATION AND AUTOMATION			
65.	SLN-2725-0000-70-S-0008 4503-KK-SG-T0000001	Instrument General Specification	A
66.	SLN-2725-0000-70-S-0007 4503-KK-SG-T0000011	Instrument Specification for Packages	A
67.	SLN-2725-0000-70-S-0018 4503-KK-SG-T0000034	General Specification - Process Connections Instrument Piping Battery Limits	A
68.	SLN-2725-5000-70-S-0011 4503-KK-SE-353T01001	Instrument Datasheet - Technical specification for Signal/Instrumentation Cable	A
69.	SLN-2725-0000-70-S-0022 4503-KK-SG-T0000035	Instrument General Specification - Symbols and Identification for Instruments Erection Drawings/Documents	0
70.	SLN-2725-0000-70-S-0092 4503-JK-SE-T000092001	Automation and Control Requirements for Packages without PLC	0
71.	SLN-2725-0000-70-W-0010 4503-KK-DT-T0000001	Instrument Hook-Up - Typicals for Process/Electrical/Pneumatic without TAG assignment	A
72.	SLN-2725-0000-70-W-0002 4503-KK-DT-T0000002	Instrument Hook-Up - Typical Instrument Earthing Diagram – Common	A
73.	SLN-2725-0000-70-S-0040 4503-JK-SE-040T01001	Technical Specification - Machine Protection System & Conditioning Monitoring System (MMS)	0

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Pos.	Document Number	Document Title	Issue
74.	SLN-2725-0000-70-S-0100 4503-JK-GS-T000000101	General Specification - Basic Plant Control and Safety Philosophy	0
75.	--	In-tools ISF format for instruments data sheet	--
76.	--	Package I&C Database format	--
77.	--	PID with project symbology / legends format	--
78.	--	Cause & effect charts format	--
		PIPING	
79.	SLN-2725-0000-55-S-0004 4503-XL-SG-T0000001	Insulation General Specification	B
80.	SLN-2725-0000-50-S-0001 4503-XZ-SG-T0000001	Layout and Piping Design Criteria	0
81.	SLN-2725-0000-50-S-0005 4503-XZ-SW-T0000001	General Specification - Equipment Welding	1
82.	SLN-2725-0000-50-S-0008 4503-XZ-SG-T0000002	General Specification - Positive Materials Identification	1
83.	SLN-2725-0000-50-S-0009 4503-XH-SL-T0000001	Piping Material Specification Index	A
84.	SLN-2725-0000-50-S-0011 4503-XH-SL-T0000003	Piping Material Classes	A

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ATTACHMENT 2 - VENDOR REQUIRED DOCUMENTS LIST (VRDL)

LEGEND:

(1) Number of Weeks after P.O. date
"$" = Documents which delayed delivery is subjected to Liquidate Damages
DEL = Delivery Time
FDS = Functional Design Specification
FAT = Factory Acceptance Test
KOM = Kick off Meeting

Note 2: for instrument data sheet, the liquidated damages are applied to In-tools forms to be provided within Delivery of equipment.

Note 3: The Liquidated Damages in case of delay in the delivery shall apply to the first valid issue of the document marked with $.
For sake of clarity, a document issued in first issue will be considered valid when it provides all information required by the Purchase Order, duly contextualized to the specific typology of the document concerned and according to all the formalized agreement (i.e. document shall be approvable with comments).
Moreover, in relation to the application of Liquidated Damages, it is agreed and understood between Parties that in case of any changes in the document's contents provided by the **VENDOR**, not specifically required by **CONTRACTOR**, in further revisions of the documents which will lead to impacts in costs and time on **CONTRACTOR**'s activities, the first issue of the document shall be considered null and void and the Liquidated Damages shall be applied.

Pos.	DESCRIPTION	With Bid	Certified for Review DUE DATE (1)	Certified for Info DUE DATE (1)	Document typology	($) Indicates Doc. Subject to Liquidate Damages
	GENERAL					
1.	Document List (Project format)			3	PL	($)
2.	Engineering, manufacturing and fabrication schedule including milestone dates for all equipment in VENDOR scope of supply.			4		
3.	Progress report showing all milestones of engineering and manufacturing, including list of order and sub-order			Monthly		
4.	Site storage, Preservation and Handling Procedure of Equipment during short and long time of non-operation			DEL-20 weeks	DS	
5.	Painting procedure along with paint product data sheets & application procedure (CONTRACTOR approval is mandatory before P.O. Issue)		16		FP	
6.	List of Sub-Vendors (CONTRACTOR approval is mandatory before P.O.)	X	4		PS	
7.	Spare Part List (CONTRACTOR form)	X		24	GP	
8.	Table of Lubricant on CONTRACTOR's Format (each equipment including drive & auxiliary items)		24		GL	

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Pos.	DESCRIPTION	With Bid	Certified for Review DUE DATE (1)	Certified for Info DUE DATE (1)	Document typology	($) Indicates Doc. Subject to Liquidate Damages
9.	Noise Source Data sheet on CONTRACTOR's Format (each equipment)		16		HN	
10.	List of Special Tools supplied with the Equipment	X		24	GT	
11.	Inspection and Test Plan (ITP) on CONTRACTOR's Format (CONTRACTOR Format is mandatory)	X	4		QC	($)
12.	Site Inspection and Test Plan	X	16		QC	
13.	Shop Inspection and Test Procedures		16		QP	
14.	Site acceptance Test Procedures	X	16		QP	
15.	Installation List / Reference List of past supplies	X				
16.	System Mechanical Data Information for Erection at Site (as per CONTRACTOR's format No. MOD-F1WI012-01)	X	6			
17.	3D Model close-out-report			3 Weeks after 30%,60 %,90% model review		
	MECHANICAL					
18.	Arrangement Drawings of equipment and auxiliary units (Plans & Elevations) such as Lube and Control Oil Units, Accessories, interconnecting piping, indicating clearance required for operation / maintenance / disassembly, including table of nozzle allowable forces & moments and nozzle thermal displacement, interface and connection schedule.	X	8		LG	($)
19.	Foundation Load (Static, Dynamic, on wind & earthquake occurrences); any unbalanced forces and moments generated by the unit in the operating speed range, position of Center of Gravity, minimum recommended weight of concrete foundation valid for piles design (+/- 10%)	X	8		LG	($)
20.	Filled-in Data sheets of each component and Performance Curves	X	6		GS	($)
21.	Piping Isometric drawings (all connection located and dimensioned), with location of vents, drains, valves, piping supports, instruments, taps etc. for piping out of the skids and for which site welding/erection is needed. (Incase any interconnecting piping are loose supplied)		16		PG	
22.	-Seal assembly Drawing and Bill of Material -Sealing System drawing of complete seal system and main components (Vessel and exchanger)			16 16	RM RD	
23.	Coupling drawings with itemized parts for spare parts identification for each sub-component (detailing sub-vendor part number)			16	RD	
24.	Piping and Instrumentation Diagram (P&ID), including but not limited to	X	6		BD	($)

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Pos.	DESCRIPTION	With Bid	Certified for Review DUE DATE (1)	Certified for Info DUE DATE (1)	Document typology	($) Indicates Doc. Subject to Liquidate Damages
	Lube oil, Process Fluid, Shaft Sealing, Machine mounted auxiliaries, Vibration & Axial Position and Temperature Monitoring systems, Cooling Water, Warm up Connections.					
25.	Utility Consumption List (State utilities needed for all normal, peak and emergency operation, fully describing circumstances of each i.e. continuous, intermittent, etc. (Air, power, water, etc.), including utilities and consumptions for Commissioning and Start-up)	X		8	BC	($)
26.	Mechanical Calculation of Pressure Vessels, Tanks, Heat Exchangers, Piping and Piping Support including relevant Fabrication / Construction Drawing		16	16	SC	
27.	Welding Book (WPS, PQR and reference Drawings)		16		FW	
28.	Equipment nameplate drawings		20		SD	
29.	Cross Section Drawing with itemized parts for spare parts for each equipment/components in the supply including, shaft seal, motor, coupling, accessories etc.			16	RD	
30.	Transportation Drawings			DEL-12 weeks	DT	
31.	Indexes of Manufacturing Data Record book (One index, common for all items)		DEL-12 weeks		DL	
32.	Indexes of Installation & Operating and Maintenance Manual (One index, common for all items)		DEL-12 weeks		OL	
33.	Indexes of Certification dossier (One index, common for all items)		DEL-12 weeks		ML	
34.	Installation Manual & Operating and Maintenance Manual complete with: - Unit operating manual giving instructions for start-up, operation, shutdown and maintenance - Recommendations for transportation and storage - Recommendations for maintenance and trouble shooting - Description of components having heavier weight and larger size for assembly and maintenance, specifying the requested tools - Erection, pre-commissioning and commissioning procedure - Manuals for instrumentation assembly, calibration, maintenance			DEL-4 weeks	OM	($)
35.	Manufacturing Data Record (common for each items)			DEL-4 weeks	DB	($)
36.	Certification dossier (one for each item) including all the certificates and report applicable.			DEL-4 weeks	MB	($)
	ELECTRICAL					
37.	Electrical Consumption List for each Component on CONTRACTOR's Format, indicating total consumption (normal and peak consumption), absorbed and installed electric power for electric users	X	8		EU	($)
38.	Electric Motor, Soft Starter, Data Sheets on CONTRACTOR's format including Motor performance & starting curves		12		ES	

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Pos.	DESCRIPTION	With Bid	Certified for Review DUE DATE (1)	Certified for Info DUE DATE (1)	Document typology	($) Indicates Doc. Subject to Liquidate Damages
39.	General Arrangement and Dimensional Outline Drawings (all Motors, JB and Soft Starter) Cross sectional drawing with part list (including terminal box drawing, wiring diagram, Grounding scheme and motor name plates drawing)		12		EE	
40.	Wiring Diagrams and Circuit Diagrams of soft starter Cabinet		12			
41.	Memory Map for Soft Starter (in project format)		20			
	INSTRUMENTATION AND AUTOMATION					
42.	Instrumentation and Control (I&C) Database as per Project Form, including: - Instrument list, - I/O list and Alarm/Trip setpoint list for each equipment included in Vendor scope of supply - HW signals exchanged with DCS/SIS, - HW signals exchanged with MCC		16		IL	($)
43.	Instrument and Valves (only actuated and PSV/TRV) data sheet (INTOOLS)		20		IS	($)
44.	List of Relief Valves and Setting		16			
45.	Utility consumption for the Total Electric Power Required for the Equipment/Instrumentation of Package Unit (UPS/Non-UPS power supply) and the Total Consumption of Instr. Air.	X	10			
46.	Calculation sheet for Flow Element/Meter, Pressure Safety valves, control Valves, wake frequency, actuator sizing and Air volume tank sizing for on-off valve		16			
47.	- Location Plan for Instruments, Junction Boxes and Local Panels - JB General Arrangement with cable entry / cable glands details		16			
48.	- Interconnection and Terminal Drawings - Junction Box / Local Panel wiring diagram - Earthing/grounding connection diagram for instruments, local panels and control cabinets (if any)		16			
49.	Cable list from instrument to junction boxes / Local Panel / Control Cabinet		20			
50.	Instrument Hook-Ups (Process, Pneumatic, Electrical, Tracing) Drawings			20	IV	
51.	C&E diagram as per Project Sample format		12		BO	
52.	SIL certificates, SIL certification reports and Safety Manuals for any item involved in SIF SIL classified.		+4W after SIL assessment			
53.	Final Documentation after DCS/SIS FAT : C&E matrix, I&C Database etc.			4 after FAT		

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Pos.	DESCRIPTION	With Bid	Certified for Review DUE DATE (1)	Certified for Info DUE DATE (1)	Document typology	($) Indicates Doc. Subject to Liquidate Damages
54.	Final Documentation after commissioning : C&E matrix, I&C Database etc			4 after commis sioning		
	AUTHORITY DOCUMENTS					
55.	Export Documentation (Technical Passports, TR CU, Certification Matrix and so on) as per project procedure					
	EXPORT CONTROL AND INTERNATION SANCTIONS					
56.	Export Compliance Details		DEL-25 weeks		EC Details form	

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ATTACHMENT 3 - VENDOR DOCUMENTS DESCRIPTION

Pos.	DOCUMENT DESCRIPTION	DEFINITION / CONTENTS	For OWNER submission
GENERAL			
1.	Document List (Project format)	List, on Project Format, to contain all documents/drawings to be submitted by the VENDOR with the following, but not limited to, details: ▪ VENDOR and CONTRACTOR document/drawing numbers ▪ Drawing description ▪ Planned submittal date ▪ Equipment number Refer also to CONTRACTOR's "Instruction for VENDOR's intermediate and Final documentation" and "Document management procedure" Where the P.O. requires the issue of a monthly report, this list shall be included and updated after first submission The native format electronic document (.XLS) shall be also submitted.	
2.	Engineering, manufacturing and fabrication schedule including milestone dates for all equipment in VENDOR scope of supply.	Document shall include detailed planning indicating the manufacturing and fabrication steps, critical points, etc. The document shall also include an indication of the first date of issue of all VENDOR documentation required for production and the maximum date by which it must be approved by CONTRACTOR without impacting the delivery date.	
3.	Construction, Pre-commissioning and Commissioning Schedule with activity break down, manpower and equipment/tools requirements	The document shall include the activities break down schedule, the manpower and equipment/tools requirements as well as activities where VENDOR's assistance may be required, specifying: ▪ Plan of activities where VENDORS assistance may be required (i.e., unpacking and check of delivery, check of installation on foundation of main and auxiliary equipment, check of temporary alignment, grouting, etc.) ▪ Estimation of expected site mobilization (Man/Weeks) ▪ Pre-Requisites for VENDOR's Intervention (i.e., foundations completed)	
4.	Progress report showing all milestones of engineering and manufacturing, including list of orders and suborders	Monthly progress report shall include: ▪ Milestones of engineering and manufacturing ▪ Fabrication schedule indicating planned & actual status ▪ Drawing list indicating planned & actual status ▪ Suborder list indicating planned & actual status: to include all sub-orders to be placed by VENDOR.	For information after internal validation of the document content (document transmission to

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Pos.	DOCUMENT DESCRIPTION	DEFINITION / CONTENTS	For OWNER submission
		Against each entry, VENDOR shall indicate anticipated award date and the latest date by which sub-order must be placed to meet the overall schedule. VENDOR shall submit unpriced copies of sub-order at the time of order placement. Where the P.O. requires the issue of a monthly report, this list shall be included and updated after first submission ▪ Vendor actions completed the previous month ▪ Vendor actions for the coming month ▪ Any corrective actions to be taken	be coordinated with post-order)
5.	Site storage, Preservation and Handling Procedure of Equipment during short and long time of non-operation	Document shall include the following details: ▪ Details as how to store equipment prior to use such that the equipment will not be damaged during storage and may enter service at a later date ▪ Actions to be done on a regular basis to maintain equipment condition Specific requirements for Warehouse storage preservation, preservation for long term storage and preservation for installed equipment to be provided. Any special unpacking/handling requirements shall be stated.	
6.	Painting procedure along with paint product data sheets & application procedure (CONTRACTOR approval is mandatory before P.O. Issue)	This document shall contain the following appropriate data as a minimum: ▪ Surface treatment with reference to required painting standard (example: EN ISO-8501, part 1) ▪ Painting- and final coating types and makes used. ▪ Number of layers and dry film thickness (DFT) of each layer. ▪ Appliance of painting and coating. ▪ Tests to be executed to prove correctness of systems used.	Yes, for review/approval
7.	List of Sub-VENDORs (CONTRACTOR approval is mandatory before P.O.)	List to include proposed SUPPLIERS of all equipment and components for package units	
8.	Spare Part List (CONTRACTOR form)	List shall indicate parts recommended by VENDOR and shall be defined by reference to cross sectional drawings and relevant parts list. Against each entry, price and delivery shall be indicated. The list shall cover all the spare parts categories such as commissioning and start-up, operational and capital spares.	Yes, for review/approval (document transmission to be coordinated with procurement)
9.	Table of Lubricant on CONTRACTOR's Format (each equipment including drive &	The lubricant list shall provide the following details for every lubricant point within the VENDOR scope:	

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Pos.	DOCUMENT DESCRIPTION	DEFINITION / CONTENTS	For OWNER submission
	auxiliary items)	▪ Lubricant point description ▪ Lubricant make (note 1) ▪ Lubricant model (note 1) ▪ Vendor to clearly indicate the International applicable nomenclature of permitted oil confirming that above make and model are not mandatory (ISO VG, SAE etc.) ▪ Lubricant first fill volume ▪ Lubricant consumption rate ▪ Flushing lubricant model ▪ Flushing lubricant filling volume ▪ Lubricant frequency of change details To note that lubricants are grease and oils. Note 1: minimum 3 oil brands shall be indicated Note 2: List shall also include in separate sheet quantities and characteristics of barrier fluid if any	
10.	Noise Source Data sheet on CONTRACTOR's Format (each equipment)	The following information shall be provided for each equipment: ▪ Equipment noise source(s) ▪ Sound pressure level (A - weighted) at 1 meter from the source in frequency bands ▪ Sound power level (A -weighted) of each source. In frequency bands.	
11.	List of Special Tools supplied with the Equipment	List shall indicate the following items: ▪ Special tools necessary for removing equipment from transport at site ▪ Special tools necessary for the installation and maintenance of the equipment ▪ Special tools necessary for commissioning and Start-up ▪ Special vendor designed tools required for the installation & maintenance of the equipment ▪ Description of each tool & function of tool ▪ Drawing or sketch of each tool	Yes, for information
12.	Inspection and Test Plan (ITP) on CONTRACTOR's Format (CONTRACTOR Format is mandatory)	Document shall include: ▪ Detailed inspection and test plan including all Quality Control steps for all involved parties (VENDOR / OWNER/ CONTRACTOR) ▪ Witness points (VENDOR / OWNER / CONTRACTOR) ▪ Hold points (VENDOR / OWNER / CONTRACTOR) ▪ Monitoring points (VENDOR / OWNER / CONTRACTOR) Final signed version of ITP shall be included in the MDR	Yes, for review/approval

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Pos.	DOCUMENT DESCRIPTION	DEFINITION / CONTENTS	For OWNER submission
13.	Site Inspection and Test Plan	Document shall include detailed inspection and test plan steps for site activities, including any VENDOR attendance.	
14.	Shop Inspection and Test Procedures	Document shall include VENDOR's procedures to illustrate compliance with specified CONTRACTOR's requirements related to inspection, covering all the tests included in the ITP, such as hydrotest, non-destructive examination, MRT, performance, etc.	
15.	Pre-commissioning and Commissioning Detailed Procedures	Document shall include VENDOR's detailed procedures to illustrate the pre-commissioning and commissioning activities for the package.	
16.	Start-up Procedure	Document shall include VENDOR's detailed procedure to illustrate the start-up activities for the package.	
17.	Site acceptance Test Procedures	Document shall include VENDOR's procedures to illustrate compliance with specified CONTRACTOR's requirements and meets process warrantees. Procedures shall indicate the following: ▪ Test bed arrangements ▪ Procedures to be adopted ▪ Readings to be taken ▪ Instruments to be used ▪ Method of interpreting readings taken to determine basis for acceptance of results ▪ If test driver/auxiliaries or job procured driver/auxiliaries is used for the test ▪ Test process medium used for the test and provide details on how this will be related to the actual process medium ▪ Acceptance points	
18.	Installation List / Reference List of past supplies.	List to include identical/similar equipment running under identical/similar service conditions delivered, installed, and operated on previous projects demonstrating good operation and reliability.	
19.	System Mechanical Data Information for Erection at Site (as per CONTRACTOR's format No. MOD-F1WI012-01)	VENDOR shall complete the form for each separately installed item of equipment or skid. Information shall be submitted for each design change affecting weight data and at the following stages: ▪ With enquiry ▪ 6 weeks after order (not exceeding values) ▪ As "final"	
MECHANICAL			
20.	Arrangement Drawings of equipment and auxiliary units (Plans & Elevations) such as Lube and Control Oil Units, Accessories, Seal System Unit, interconnecting piping, Oil Overhead Tank, indicating	Envelope and dimensions relative to unit datum. Plant North arrow to be shown. Access, withdrawal, and laydown requirements for maintenance to be shown. Location and numbering (as per project methodology) of all piping and tubing terminations for CONTRACTOR connections.	Yes, for review/approval

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Pos.	DOCUMENT DESCRIPTION	DEFINITION / CONTENTS	For OWNER submission
	clearance required for operation / maintenance / disassembly, including table of nozzle allowable forces & moments and nozzle thermal displacement.	Identification and location of all major on skid components with CONTRACTOR's tag numbers added. Overall weight and maintenance weights for major components, Center of Gravity, lifting points and spreader beam to be shown. Drawing to indicate acceptable loads, forces and moments on flanges to which CONTRACTOR connects. Equipment thermal expansion requirements/information, where applicable, shall be clearly reported in the drawing. Foundation footprint requirement for items installed on concrete pedestals shall also be shown. Where a package consists of more than one skid, separate drawing shall be provided for each skid, over the general Arrangement Drawing. General Arrangement shall also include a general 3D view of equipment.	
21.	Foundation Load (Static, Dynamic, on wind & earthquake occurrences); any unbalanced forces and moments generated by the unit in the operating speed range, position of Center of Gravity, minimum recommended weight of concrete foundation; Anchor Bolt Sizes, Location & Projection, Grouting Details	Foundation load shall include: • Weight of each item on skid (equipment, driver, gear box, base plate, etc.), and overall weight of the assembly. • Operating speeds (RPM) of Motor and Pump and/or gear box ratio. • Centre of gravity of equipment, driver, gearbox and combined unit (in plan and elevation). • Rotor masses for equipment and driver. • Dynamic load & unbalanced forces (if any). If unbalance forces are negligible, the vendor should indicate the same in the drawing that "Dynamic loads are negligible". For reciprocating machine, the vendor shall provide details of application of unbalance forces and torques about both the horizontal axis and vertical axis and its location w.r.t. reference point of the skid. • Scope of supply of Anchor bolts. If anchor bolts are supplied by Vendor, the information about size, location of anchor bolt pocket and depth of pocket. In case anchor bolts are not supplied by vendor, the type, material and capacity of the anchor bolt shall be as per project standard anchor bolt drawing. Vendor shall use the capacity indicated for working out number of anchor bolts. Anchor bolt location shall be provided w.r.t. centre line of discharge to locate the item correctly in layout • Type of grout. 'Non-Shrink Cement Grout' below skid is by default unless there is specific requirement. The extent of grout to be provided below the skid.	Yes, for review/approval

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		• Allowable amplitude and/or vibration velocity at top of foundation. • Requirement of base plate/skid frame filling. (we need to provide screed only if required specifically by vendor) • Any accidental loads to be considered for static design of the foundation. Where a package consists of more than one skid, separate drawing shall be provided for each skid, over the general Arrangement Drawing. Drawing may be combined with position 20 "Arrangement Drawings of equipment and auxiliary units"	
22.	Filled-in Data sheets of each component and Performance Curves: VENDOR shall submit API Data Sheet dully filled in.	Where the Equipment Data Sheets are issued by CONTRACTOR as part of Purchase Order, VENDOR to fully complete in all its part. Where Data Sheets are not issued by CONTRACTOR, VENDOR to use an approved format.	Yes, for review/approval
23.	Interface and Connection Schedule	Listing by number of all VENDOR termination points, including electrical and instrument cable terminations and all junction boxes cross referenced to the relevant drawings. Size, rating and specification/reference code of all piping and tubing termination points requiring CONTRACTOR connection. Identification of corresponding connection point on another skid or system to which each point shall be connected. Identification of fluids at each connection point including pressure and temperature conditions. For each vent and drain, fluids under normal and abnormal operating conditions shall be stated, and system to which each must be connected (i.e., CONTRACTOR's vent or drain – safe, open or closed). Drawing may be combined with position 20 "Arrangement Drawings of equipment and auxiliary units" and shall be always submitted together with the General Arrangement Drawing.	
24.	Equipment List (common for all items)	A list containing the major equipment data. In case of a CONTRACTOR's or OWNER's form shall not be used for the Equipment List, the following information is required: ▪ CONTRACTOR/OWNER Equipment tag number ▪ Equipment description ▪ Material in contact with process ▪ Major duty details - for pumps, flow and head; for compressors, flow and pressure; for heat exchangers, flow and heat transfer or heat transfer area; for filters, surface area; for vessels, size	

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Pos.	DOCUMENT DESCRIPTION	DEFINITION / CONTENTS	For OWNER submission
		▪ Equipment major size - for vessels, internal width, height, diameter ▪ Equipment required power ▪ Design and Operating pressures ▪ Design and Operating temperatures ▪ Pressure test value ▪ Pressure test fluid ▪ ATEX classification ▪ Area number ▪ P&ID number ▪ Coating / Insulation / Fireproofing data Shipped loose material/items requiring site installation and assembly shall be clearly identified	
25.	Line List	For the interconnecting piping, where the CONTRACTOR/Project standard line list is not used, this document shall contain the following data: ▪ Line number ▪ Location from ▪ Location to ▪ Process medium ▪ Process phase ▪ Operation temperature ▪ Operation pressure ▪ Design pressure ▪ Design temperature ▪ Test pressure ▪ pressure test fluid ▪ Material in contact with process ▪ Flow medium hazard category ▪ Insulation data Tracing data to be provided in a spreadsheet (.XLS) format	
26.	Piping Isometric drawings (all connection located and dimensioned), with location of vents, drains, valves, piping supports, instruments, taps etc.	Isometric drawings of the piping in the package. Piping isometrics shall include the following details: ▪ Piping dimensioned (lengths and angles) isometric view ▪ Flange connections (and location dimensions) ▪ Welded connections (and location dimensions) ▪ Field welds (and location dimensions) ▪ Support locations and support types at these locations (fixed location, sliding location) ▪ Plant north arrow ▪ Piping identification number / line number / spool number relating spool as delivered to the drawing ▪ Material take-off ▪ All inline components ▪ All instrument connections ▪ Flow arrow ▪ X,Y,Elevation co-ordinates	

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Pos.	DOCUMENT DESCRIPTION	DEFINITION / CONTENTS	For OWNER submission
		Specific case where CONTRACTOR/Project standard template to be applied, the same will be agreed between the parties during VDL definitions	
27.	Seal assembly Drawing and Bill of Material	The drawing shall include: ▪ Model no. and size ▪ Main dimensions and clearance including location of gland plate ▪ Manufacturer's material code ▪ API 682 mechanical seal classification code ▪ Hard facing under flexible members and auxiliary packing ▪ Parts list showing materials of construction ▪ Operating conditions, including equipment chamber pressure ▪ API 682 seal flush, injection, quench and vent connections ▪ Quantity for seal flush, injection and quench ▪ Plugged connections to be identified ▪ Cross-section of seal plate, showing flush, quench etc. holes ▪ Direction of rotation ▪ Direction of spring winding	
28.	Sealing System drawing of complete seal system and main components (Vessel and exchanger).	Document shall as minimum include: ▪ Arrangement Drawings (Plans & Elevations) of complete seal system with stand (if applicable); 3D view shall also be included; ▪ Bill of material detailing as minimum for each component size, quantity, material, rating, end finishing. ▪ Weights (empty and operating); ▪ Barrier fluids data (type, quantities etc.); ▪ Refill/Charging data;	
29.	Coupling drawings with itemized parts for spare parts identification for each sub-component (detailing sub-vendor part number).	Document shall be also complete of duty conditions, technical and sizing conditions/calculations of the coupling itself, balance detail.	
30.	Piping and Instrumentation Diagram (P&ID), including but not limited to Lube oil, Process Fluid, Shaft Sealing, Machine mounted auxiliaries, Vibration & Axial Position and Temperature Monitoring systems, Cooling Water, Warm up Connections.	Piping and Instrumentation diagrams (P&ID's) shall be provided for the whole scope of delivery. The P&ID's shall be drawn by VENDOR using standard symbols provided on CONTRACTOR's legend sheet. P&ID is the representation of all itemized equipment, including stand-by equipment, process/utility lines with preliminary sizes and recommended materials, valves, tracing requirements, process instruments, special process requirements, such as required elevation and slope. In detail the P&ID's shall include:	Yes, for review/approval

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Pos.	DOCUMENT DESCRIPTION	DEFINITION / CONTENTS	For OWNER submission
		• Scope boundary lines clearly showing demarcation between VENDOR and CONTRACTOR scope of supply • Battery limit connections: Process, Utilities, Auxiliary including drain point (e.g. funnels, low point valves); with identification numbers, size, rating and functional description. • All process and utility equipment, including spares. • Equipment name item numbers including equipment's drivers. • Elevations of major equipment. • Vessels Nozzle and Slope indication. • All process and utility flowlines with directional arrows (and, if required for clarity, also outside the vendor scope). • Line sizes, numbers, pipe specifications, specification breaks, product designation and Piping notes (e.g. Slope, Do Not Pocket). • Insulation and trace heating requirements (electric, steam, hot, cold, personal protection, acoustic). • Venting and draining including funnel indication • All indication and controls required for safe operation of the equipment. • Relief requirements - PSVs location set pressures tag numbers and sizes. • Instruments and switches with tag numbers, and alarm trip set points • Valves with sizes, actuators and solenoids. Failure mode. P&ID's may either be separate i.e. for equipment, lube oil, cooling water, safety (vibration, etc), seal system or combined as one common P&ID.	
31.	Utility Consumption List	This document contains the characteristics of all utilities used (electrical consumptions are excluded as covered by position 46). The list specifies the consumption for all normal, peak, start-up, shutdown and emergency operation, fully describing circumstances of each i.e. continuous, intermittent, etc. This document shows a list with utilities consumption such as: ▪ Water ▪ Steam ▪ Condensate ▪ Fuel gas ▪ Nitrogen	

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Pos.	DOCUMENT DESCRIPTION	DEFINITION / CONTENTS	For OWNER submission
		▪ Plant/instrument air ▪ Etc.	
32.	Mechanical Calculation of Pressure Vessels, Tanks, Heat Exchangers, Piping and Piping Support including relevant Fabrication / Construction Drawing	Stress calculations shall be in accordance with relevant code requirements and demonstrate that design (including nozzles) is adequate for operation within the parameters specified for the item in terms of pressure, temperature, nozzle loadings, etc. Also to include calculations for lifting lugs and brackets. The relevant fabrication / construction drawing, shall be presented as part of and together with the calculation, showing identical data as used for the cross-reference identification to the calculation.	
33.	Lateral critical speed calculations	Calculations shall determine the natural frequency of the shaft assembly and identify forcing frequencies and harmonic components thereof, relative to operating speed range. Results shall be presented in graphical and narrative form, and shall include: ▪ Rotor drawings showing each shaft segment clearly numbered ▪ Table of masses and stiffness values for segment ▪ Plot of critical speed against support stiffness with stiffness both in the vertical plane and horizontal plane to be shown for each support point ▪ Plots of speed against vibration amplitudes (Campbell diagrams) to demonstrate the specifications of operating modes form harmonic regions	
34.	Torsional critical speed calculations	Calculations shall determine the torsional critical speeds for driver/transmission/driven equipment train. Calculations shall clearly indicate number and details of finite elements that the system has been divided into for the calculation and a table of stiffnesses and inertias for each element shall be included. Results shall be presented in graphical (Campbell diagrams) and narrative form.	
35.	Welding Book (WPS, PQR and reference Drawings)	Welding Book shall contain: ▪ WPS (Welding Procedure Specification) produced by the VENDOR describing the geometry of the weld, material to be used, etc. ▪ PQR (the WPS qualification) ▪ Sketch or drawing indicating all weld numbers with reference to WPS and/or numbers used for NDT (this may be an existing detail or GA Drawing) Fabrication shall not commence before the appropriate weld procedure has been approved by CONTRACTOR.	
36.	Equipment nameplate drawings	Nameplate drawing shall indicate at least the following: ▪ Nameplate data	

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Pos.	DOCUMENT DESCRIPTION	DEFINITION / CONTENTS	For OWNER submission
		▪ Nameplate dimensions ▪ ATEX reference as necessary ▪ All necessary stamps Nameplate shall include the information/parameters required by the equipment standard and/or Project Specifications (i.e., API, ISO, etc.). Each equipment shall be identified with a univocal serial number from 1st revision. Where double language is required, nameplate drawing shall be issued accordingly from 1st revision.	
37.	Cross Sectional Drawing with itemized parts for spare parts identification for each equipment / component in the supply including, shaft seal, motor, coupling, accessories etc.	Drawing for all main subcomponents, indicating the following details: ▪ Cross sectional drawing of equipment ▪ Index of parts (parts list) ▪ Equipment internal details such as: impellers for pumps / compressors, filter elements and filter elements sealing for filters, column packing and tray details, heat exchanger bundle details, mixer impeller details ▪ Gasket and sealing surfaces and materials ▪ Materials list ▪ Part size / dimensions / make / model	
38.	Lube oil and sealing system component data	If different than position 22, “Filled-in Data sheets of each component”.	
39.	Transportation Drawings	These drawings shall show as a minimum: ▪ Size of equipment (or parts thereof) ▪ Weight of equipment (or parts thereof) ▪ Details on how to lift / move the equipment (or parts thereof) ▪ Size, number, weight, identification and contents of transportation boxes / containers ▪ Any special unpacking / handling and shipping requirements	
40.	Indexes of Manufacturing Data Record book (One index, common for all items)	Refer to dedicated Project Procedure	
41.	Indexes of Installation & Operating and Maintenance Manual (One index, common for all items)	Refer to dedicated Project Procedure	
42.	Indexes of Certification dossier (One index, common for all items)	Refer to dedicated Project Procedure	
43.	Installation Manual & Operating and Maintenance Manual	Refer to dedicated Project Procedure	
44.	Manufacturing Data Record (common for each items)	Refer to dedicated Project Procedure	
45.	Certification dossier (one for each item) including all the certificates	Refer to dedicated Project Procedure	

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Pos.	DOCUMENT DESCRIPTION	DEFINITION / CONTENTS	For OWNER submission
	and report applicable.		
ELECTRICAL			
46.	Electrical Consumption List for each Component on CONTRACTOR's Format, indicating total consumption (normal and peak consumption), absorbed and installed electric power for electric users	The electrical consumption list shall include: ▪ Equipment tag number ▪ Electrical Load description ▪ Electrical Load type (heater, motor, instrumentation) ▪ Supply voltage ▪ Supply phase ▪ Supply power rating (kW)	
47.	Electric Motor Data Sheets on CONTRACTOR's format including Motor performance & starting curves	Motor data sheet shall include: ▪ Equipment tag number ▪ Motor make ▪ Motor model ▪ Motor power rating ▪ Efficiency ▪ Power factor ▪ Full load current ▪ Starting current Ia / In ▪ Power factor at start ▪ EEx classification ▪ IP rating ▪ Insulation class ▪ PTC/PT100 elements installed y/n ▪ Anti-condensation heater installed y/n ▪ Speed / torque starting characteristics	Yes, for review/approval
48.	Dimensional Outline Drawings with part list (including terminal box drawing, wiring diagram and name plate drawings)	Drawings shall indicate the electrical arrangements of all component parts, including cable type and color.	
49.	Motor Cross Sectional Drawing with Part List (for main motor)	Drawing for all main subcomponents and shall indicate the following details: ▪ Cross sectional drawing of equipment ▪ Index of parts (parts list) ▪ Equipment internal details ▪ Gasket and sealing surfaces and materials ▪ Materials list ▪ Part size / dimensions / make / model	
INSTRUMENTATION AND AUTOMATION			
50.	Instrument list and Point Configuration Data (PCD) as per CONTRACTOR Project Samples format	The instrument list shall contain the following information as a minimum: ▪ Listing of all instruments ▪ Tag number ▪ Instrument description (pressure switch, control valve, level gauge, etc.) ▪ Service description ▪ I/O with signal type for each equipment included in Vendor scope of supply ▪ I/O with signal type for interface with DCS/SIS	

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		▪ I/O with signal type for interface with MCC ▪ Location of line (size, number, spec.) ▪ P&ID number ▪ Data sheet number ▪ Alarm, trip and shutdown setting ▪ Instrument range	
51.	Instrument and Valves (only actuated and PSV/TRV) data sheet (INTOOLS)	Instrument data sheet shall contain the following data as a minimum: ▪ Required process data and units ▪ Instrument capable process data and units ▪ Instrument make ▪ Instrument type ▪ Instrument model ▪ Instrument material of construction for process contact parts ▪ Instrument material of construction for non-process contact parts ▪ Instrument overall size ▪ Weight (only for panels, cabinets, shelters and analyser houses) ▪ Instrument ATEX classification (category, group, class) ▪ Instrument seal data (if fitted) ▪ Instrument required power / voltage level ▪ Instrument nozzle connection data, and service ▪ Instrument range ▪ Instrument controller action ▪ Instrument flow limits (if applicable) ▪ Instrument pressure limits (if applicable) ▪ Instrument temperature limits (if applicable) ▪ Loop impedance ▪ Cable gland and cable entry details ▪ Process connection details ▪ SIL classification FOR VALVES: ▪ Duty ▪ Medium ▪ Size ▪ Make ▪ Model ▪ Material	
52.	Instrument Calculation Report	Vendor shall provide calculation for the following items where applicable: - Control Valve Sizing - On/Off Valves Actuator Sizing - On/Off Valves Volume Tank Sizing - PSV's/TRV's Sizing	

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Pos.	DOCUMENT DESCRIPTION	DEFINITION / CONTENTS	For OWNER submission
		- Flow Element Sizing - Wake Frequency Stress Calculations shall be submitted for approval prior to the placement of orders for the related valves	
53.	Instrument wiring diagram including JB details as per Project Sample format	Diagrammatic drawing, as per Project sample format. Instrument cable termination details shall show junction box gland plate drilling sizes to suit external cabling to/from the package, and all glanding information. All cables indicated on these drawings must be terminated at both ends.	
54.	Instrument Hook-Ups (Process, Pneumatic, Electrical, Tracing) Drawings as per Project Sample format	Process hook-up drawings, as per Project sample format, shall be prepared for each type of instrument that requires a process impulse line for sensing purposes. Similarly, a pneumatic hook-up drawing shall be prepared for each type of instrument air transmission/control signal. Both types of drawing shall include all the necessary mounting details and a schedule of all the installation materials used.	
55.	C&E diagrams as per Project Sample format	C&E diagrams shall indicate clearly and precisely the shutdown requirements on the standard format sheet with defined convention. Individual C&E diagrams shall be produced for each process unit. All auto start/change over for each equipment shall be clearly defined with location of field devices.	
EXPORT CONTROL AND INTERNATION SANCTIONS			
56.	Export Compliance Details	As per the Export Compliance Details Form attached to the Special Purchase Conditions	

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